Ecohydrology

Long-Term Water Balance Dynamics in a Brazilian Neotropical Forest: Insights from Seven Years of Continuous Observation

Journal:	*Ecohydrology*
Manuscript ID	ECO-24-0224
Wiley - Manuscript type:	Review Article
Date Submitted by the Author:	25-Sep-2024

Complete List of Authors:	Guauque-Mellado, Daniel; Editora da Universidade Federal de Lavras Rodrigues, André Ferreira; Universidade Federal de Minas Gerais Escola de Engenharia Guo, Li; Sichuan University State Key Laboratory of Hydraulics and Mountain River Engineering Junior, José Alves Junqueira; Instituto Federal de Educacao Ciencia e Tecnologia do Sudeste de Minas Gerais - Campus Avancado Bom Sucesso Mello, Carlos; Editora da Universidade Federal de Lavras

Keywords:	soil water storage, biophysical controls, rainfall interception, evapotranspiration, Atlantic Forest

Long-Term Water Balance Dynamics in a Brazilian Neotropical Forest: Insights from Seven Years of Continuous Observation

Daniel Guauque-Mellado[1], André Ferreira Rodrigues[2], Li Guo[3], José Alves Junqueira Junior[4], Carlos Rogério de Mello[1]

[1]Water Resources Department, School of Engineering, Federal University of Lavras, CP 3037, Lavras, MG, Brazil, 37200-900; [2]Department of Hydraulics and Water Resources, School of Engineering, Federal University of Minas Gerais, CP 6627, 31270-901, Belo Horizonte, MG, Brazil; [3]State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resource and Hydropower, Sichuan University, Chengdu 610000, China;[4]Instituto Federal do Sudeste de Minas Gerais, Campus Bom Sucesso, Bom Sucesso, MG, Brazil, 37220000

Abstract

The Atlantic Forest ecosystems offer numerous ecosystem services. However, UNESCO regards it as one of the most fragile biosphere reserves globally due to anthropogenic pressures in Brazil that threaten its integrity. This study evaluated the water balance relationship in an Atlantic Forest fragment using hydrometeorological data collected over seven hydrological years (2014-2020) of continuous observation of the hydrometeorological elements and soil moisture. The focus was on analyzing evapotranspiration and its biophysical controls, the canopy rainfall interception, soil moisture, and percolation over time. On average, canopy rainfall interception accounted for 19% of gross precipitation (GP), and evapotranspiration constituted an average of 77% of GP, peaking at 90% during the driest year of the study period. Evapotranspiration processes were primarily governed by stomatal and soil water storage, mainly during dry periods. Percolation into the subsoil showed significant variation, ranging downward and upward during rainy and dry periods, respectively, driven by increased evapotranspiration and a marked reduction in GP and soil moisture. The water balance components highlighted that the Atlantic Forest is a crucial biome and a lifeline for providing ecological functions, particularly in water control and storage, even during dry periods.

Keywords: soil water storage, biophysical controls, rainfall interception, evapotranspiration, Atlantic Forest.

Introduction

The Atlantic Forest Biome, home to more than 60% of Brazil's urban population, is under considerable pressure and fragmentation. This degradation has led to vegetation losses and adverse impacts on climate and hydrological regimes, underlining the urgent need for conservation. Taffarello et al. (2017) and Macedo et al. (2021) have emphasized the crucial role of Atlantic Forests in providing water services, mitigating persistent droughts, and reducing natural hazards in Brazil. UNESCO has recognized the Atlantic Forest as one of the most important biosphere reserves worldwide, given its diversity, endemism, and environmental significance (Myers et al., 2000). Currently, the biome retains only 28% of its original coverage, with 91% of this area experiencing continuous fragmentation. In addition, 46% of the remaining forest is subject to a strong edge effect (Macedo et al., 2021; Poorter et al., 2016) and exhibits low resilience (Tambosi et al., 2014). These facts highlight the pressing need to preserve the remaining biodiversity and ecohydrological functions of the Atlantic Forest Biome (Silva et al., 2020; Rezende et al., 2018). Studying the hydrological dynamics and processes of this ecosystem is crucial and urgent to support strategies for the conservation and sustainable management of the Atlantic Forest. Among the variables of the forest water balance, evapotranspiration is configured as one of the most important variables as it connects the soil to the atmosphere and plays a critical role in water and energy cycles (Guauque-Mellado et al., 2022; Cabral et al., 2015; Costa et al., 2010; Marques et al., 2020; Tan et al., 2019). Therefore, accurately evaluating its influence on the water balance of the Atlantic Forest is essential to enhance our understanding of the regional climate dynamics, water yield, and its quality, and improve the management of these ecosystems.

Although several studies have examined the evapotranspiration of Atlantic Forest biomes, they are mainly short-term studies, spanning less than two years (Mello et al., 2019). Hydrology in some areas of the Atlantic Forest is affected by a marked climatic seasonality, characterized by dry winters and wet summers, resulting in its semi-deciduous nature (Freitas et al., 2020). Species in semi-deciduous ecosystems typically self-regulate the physiological process of transpiration, shedding foliage during rainless periods to avoid water stress (Marques et al., 2020). This aspect should be considered in studies of forest water balance, as it can affect water storage capacity throughout the year (Mello et al., 2019). Therefore, previous studies offered only partial insights into this ecosystem's dynamics of evapotranspiration and water pathways. In addition, the semi-deciduous nature of the Atlantic Forest, marked by climatic seasonality, presents a unique challenge for understanding its water balance. The interaction between meteorological variables, vegetation composition, phenology, and soil water attributes on water balance in Atlantic Forest areas remains poorly understood. Therefore, further knowledge of evapotranspiration and the water balance partitioning in this ecosystem, based on long-term hydrometeorological measurements, is crucial for a comprehensive understanding, especially in the context of climate change.

Studies in other ecosystems have directly measured evapotranspiration flux using the Eddy Covariance system (e.g., Rocha et al., 2004; Cabral et al., 2015; Marques et al., 2020). Besides, effective methodologies also exist for indirectly calculating evapotranspiration, such as energy and water balance with residual focus (Rodrigues et al., 2021; Jones et al., 2017). Models like the large leaf or Penman-Monteith, when validated against observed datasets, can accurately capture the variability of climate and soil water storage, thereby making the water balance analysis in forest ecosystems reliable (Guauque-Mellado et al., 2022; Jones et al., 2017; Kume et al., 2011; Pereira et al., 2010).

This study aims to evaluate the water balance components of an Atlantic Forest remnant based on hydrometeorological data spanning seven hydrological years (October 2013 to September 2020). Specifically, we seek to (i) identify the dynamics and partitioning of the rainfall interception and its influences on soil water storage; (ii) estimate evapotranspiration using meteorological data and identify its main drivers; and (iii) estimate the elements of the water balance using a unique hydrological dataset that encompasses both dry and wet years.

Materials and Methods

Site description

The studied area (Figure 1) is in Southeast Brazil, in the coordinates 21º13'40''S and 44º57'50''W, with an average altitude of 925 meters. Consists of a seasonal semideciduous mountain forest (SSMF) remnant with 6.35 ha. The soil is a Red Latosol distributed over a slightly undulated relief with slopes < 15% (Junqueira et al., 2017). The Köppen climate of the region is mesothermal subtropical, i.e., Cwa, with well-defined precipitation seasonality (Junqueira Junior et al., 2019): wet (October to March) and dry (April to September) periods. The mean annual precipitation is 1461.8 mm, with 84% concentrated in the wet season. The mean annual temperature is 20.3°C, varying from 16.9°C in July to 22.8°C in February (INMET, 2022).

The experimental area has been periodically monitored regarding vegetation parameters (forest inventory) since 1994, measuring species growth and identifying tree mortality. At first, it was inventoried by 6527 individuals with a diameter at breast high (DBH) > 5 cm by Oliveira-Filho et al. (1994). However, 5609 individuals were registered in the last forest inventory (2017), distributed among forty-eight botanical families and 184 species. The most representative families are *Annonaceae (830), Fabaceae (769), Lauraceae (720), Rubiaceae (671), Myrtaceae (510), Meliaceae (431), and Melastomataceae (363).* The most frequent species are: *Xylopia brasiliensis* (670)*, Copaifera langsdorffii* (517)*, Amaioua intermedia* (497)*, Trichilia emarginata* (383)*, Miconia willdenowii* (282)*, Myrcia splendens* (262)*,* and *Casearia arbórea* (195).

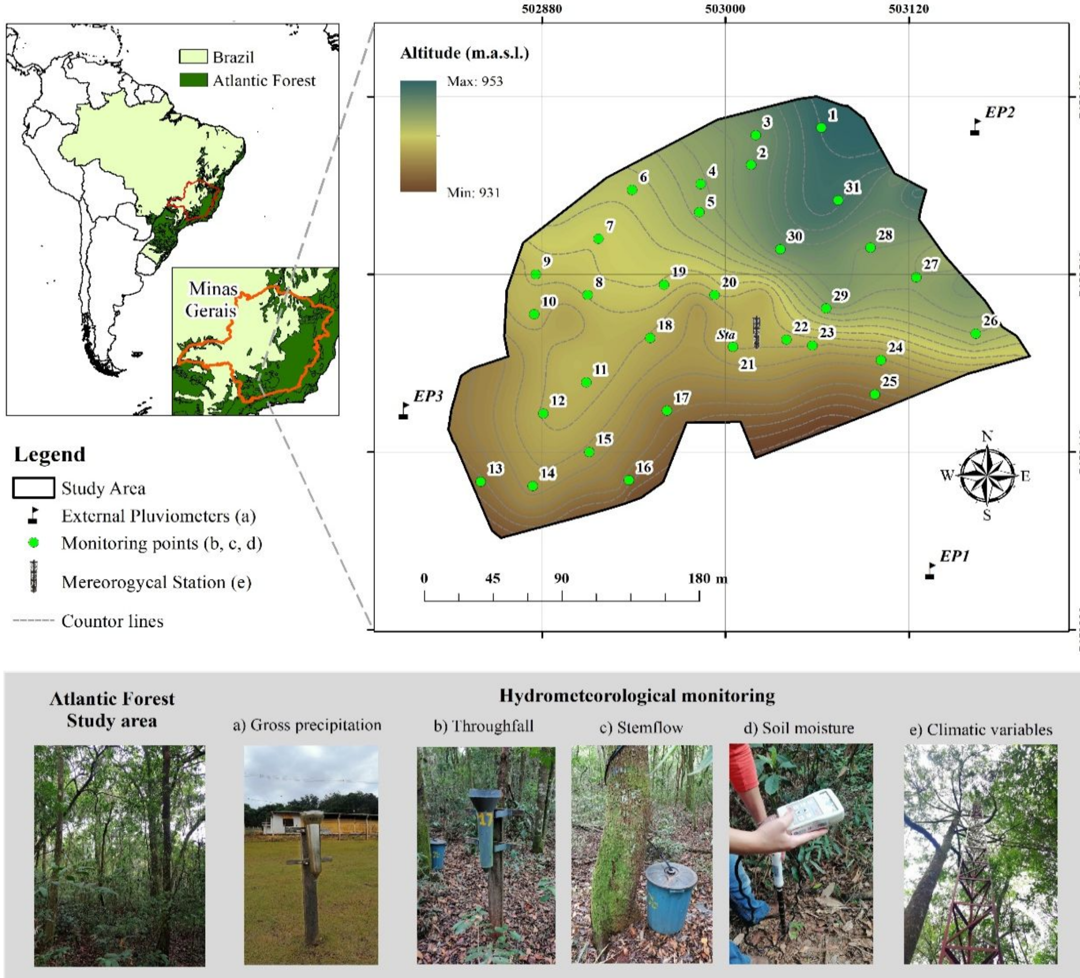


Figure 1. Geographical location and instrumentation used for hydrometeorological monitoring in the SSMF, southeastern Brazil.

The forest is in a stage of transition (Rodrigues et al., 2021), with 87.68% of individuals in minor classes of DBH, i.e., 5-15 cm (68.41%) and 15-25 cm (19.29%). The other classes correspond to 12.32%, in which the 25-35 cm class comprehends 8.34% and > 35 cm, 3.97%. The forest canopy's average height is 10.2 m, with a few dispersed individuals higher than 20 m (Junqueira et al., 2019). The canopy stratification has four levels: (i) a layer of emerging trees (>20 m) dispersed in the area; (ii) a layer of the main canopy (10-15 m); (iii) a layer of shrub and young trees (<10 m); and (iv) an herbal cover over the soil. Openings in the canopy are caused by fallen and dead trees that have completed their cycle, which modifies the species' dynamic and the remnant's structure.

Monitoring framework

Gross precipitation (GP), i.e., the incoming rainfall, was measured between October 2013 and September 2020. In this period, 644 rainy events were monitored. At first, between 2013 and 2017, the measurement was carried out through two procedures: (i) automatic monitoring, performed every hour using an 8-inch tipping-bucket rain gauge that covers 324.3 cm^2 (TE525-L, Texas Electronics Inc., Dallas, TX, EUA); and (ii) daily manual monitoring, performed using a totalizing-type "Ville de Paris" rain gauge with a collected area of 378.5 cm^2. Both rain gauges were installed at a 22 m high Micrometeorological Observation Tower (MOT) near the study area's center (Figure 1). From August 2017, three other totalizing-type rain gauges were installed surrounding the SSMF to complement the monitoring system and improve the spatial representativeness of GP (Figure 1a).

In the MOT (Figure 1e), micrometeorological sensors were installed at two heights, one at the top of the forest canopy (22 m high) and another at 2 m above ground. Each level contains sensors for relative humidity (RH) and temperature (T_a) measurements (Model S-THB-M002). A pyranometer set (measuring incoming and outgoing radiation) (Model S-LIB-M003) for net radiation (Rn), a barometric pressure sensor (Model CS100), and an anemometer (Model 03002) for wind speed and direction were also installed. The sensor data were stored hourly in a CR10x programmable data logger (Campbell Scientific, Logan, UT, USA).

Thirty-one points were randomly selected inside the SSMF to measure throughfall (TF), stemflow (Stf), and soil moisture (θ) up to 1.0 m depth (Figure 1). These points are approximately spaced 40 m from each other. TF measurement (Figure 1b) was carried out with a "Ville de Paris" rain gauge installed 1.5 m above the forest floor to avoid splash-in (Junqueira et al., 2019). Data collection was performed four hours after the rainfall had ceased (and on the following day when it ceased late afternoon) to guarantee that the canopy had wholly drained and prevented bias.

Sf was measured in one representative tree (Figure 1c) for each monitoring point. The individuals were selected according to localization, abundance, and dominant species (Terra et al., 2018). The collectors were constructed with polyethylene hose set in the tree trunks in a spiral scheme, applying silicone between the collector and tree to avoid leaking (Mello et al., 2023). The stemflow was drained into a container and collected simultaneously to TF.

In each monitoring point, permanent access tubes were installed (Figure 1d) to measure the soil moisture ($m^3 m^{-3}$) in five depths (0.10 m, 0.20 m, 0.30 m, 0.40 m, and 1.0 m) at ~ 15-day intervals. The measurement was carried out with a multi-sensor capacitance probe (model PR2/6, Delta-T Devices Ltd.).

The leaf area index (LAI in $m^2\ m^{-2}$) was measured monthly at the 31 monitoring points using a previously calibrated and validated Sun Scan zeptometer system (Model SS1-STD3, Delta-T Devices Ltd., Cambridge, UK). These LAI datasets were used to parametrize the heat flux from the soil considered in the Penman-Monteith model for evapotranspiration estimates.

Water balance calculation

The water balance of the forest remnant was computed according to Jones et al. (2017):

$$\frac{\Delta S}{\Delta t} = \text{GP-ET- C-DP} \tag{1}$$

Where GP (mm) is the gross precipitation, ET (mm) is the evapotranspiration, C (mm) is the canopy interception, S (mm) is the soil water storage integrated over the root zone (1.0 m in this study), DP (mm) is the deep-water percolation leaving the root zone, and Δt is the interval adopted for the water balance computation (monthly). GP was calculated by averaging the precipitation of the external rain gauges. $\frac{\Delta S}{\Delta t}$

was calculated for each one of the 31 sets of monitoring as a function of the soil moisture readings interval. Although tropical forests can uptake water from deeper layers during drought conditions (Markewitz et al., 2010), this amount is of minor contribution to total evapotranspiration as it is restricted to short periods (Broedel et al., 2017). Thus, the root water uptake was considered up to 1.0 m depth since we observed a decrease in thinner root abundance during the installation of the monitoring system, concentrated between 0.50 and 1.0 m. S was calculated as follows:

$$S = \sum_{i=1}^{n}\left(\frac{\theta_i + \theta_{i+1}}{2}\ x\ h\right) \tag{2}$$

$$\frac{\Delta S}{\Delta t} = \frac{S_{(j+t)} - S_{(t)}}{\Delta T} \tag{3}$$

where θ_i e θ_{i+1} are, respectively, soil moisture at depth i and i + 1($m^3\ m^{-3}$), n is the number of layers (0–0.10 m; 0.1–0.2 m; 0.20 –0.30 m; 0.30–0.40 m; 0.40–1.0 m), h is the thickness of each layer (mm), j is the water balance time step and t is the consecutive date of each measurement.

The evapotranspiration can be calculated using the water balance equation (ET_{WB}) in periods of low rainfall, i.e., when percolation can be neglected (Tomasella et al., 2008; Mello et al., 2019):

$$\frac{ET_{(WB)}}{\Delta T} = \frac{(TF + Stf) - \Delta S_{(0\text{-}1.0\ m)} + C}{\Delta T} \tag{4}$$

where C is the average canopy rainfall interception (mm) (using the 31 observed points), obtained through the equation described by Ghimire et al. (2014) and Junqueira et al. (2019):

$$\frac{C}{\Delta T} = \frac{GP - (TF + Stf)}{\Delta T} \tag{5}$$

Percolation was estimated as a residue of the water balance from Equation 1 in the rainy period. According to Jones et al. (2017), this simplification implies that when GP exceeds ET, P tends to be positive, meaning that the water is leaving the root zone, i.e., percolation. On the other hand, when ET exceeds GP, P is negative, and water enters the root zone (capillary rise).

174 Since forest evapotranspiration relies on soil water availability, we calculated the relative
175 extractable water (REW) (Allen et al., 1998):

176 $$REW = \frac{(\theta - \theta_{wp})}{(\theta_{fc} - \theta_{wp})} \quad (6)$$

177 where θ, θ_{fc}, and θ_{wp}, are the water volumetric content, the field capacity, and the wilting point, that were
178 obtained by Rodrigues et al. (2020).

179 *Parameterization of the evapotranspiration model*

180 During the rainy season (October – March), the dependent variable of the water balance was the
181 deep percolation (DP), with equation 2 rewritten as:

182 $$\frac{DP = GP - ET_M - C - \Delta S}{\Delta t} \quad (7)$$

183 We modeled ET for SSMF using equation 8. For that, it was also performed a parameterization of
184 the ET encompassing soil evaporation and plant transpiration into a single evaporative element process
185 (Ershadi et al., 2015), which is used to describe the physiological and environmental conditions (Monteith,
186 1965; Kelliher et al., 1995). ET is estimated by:

187 $$ET = \frac{\Delta \cdot (R_n - G) + \rho_a \cdot c_p \cdot g_a \cdot (VPD)}{\lambda \cdot \left[(\Delta + \gamma) \cdot \left(1 + \frac{g_a}{g_s}\right)\right]}$$

188 (8)

189 which ET is the modeled evapotranspiration (mm d^{-1}); λ is the latent heat of water vaporization (MJ kg^{-1});
190 Δ is the declivity of saturation vapor pressure curve (kPa $°C^{-1}$); R_n is the net radiation (MJ m^{-2} d^{-1}); G is the
191 heat flux in the soil (MJ m^{-2} d^{-1}); γ is the psychrometric coefficient (kPa $°C^{-1}$); ρ_a is the air density (kg m^{-3});
192 VPD is the water vapor pressure deficit (kPa); Cp is the specific heat at a constant pressure (J kg^{-1} $°C^{-1}$)
193 (approximately 1,000 J kg^{-1} $°C^{-1}$); g_a is the aerodynamic conductance (m s^{-1}), and g_s is the canopy
194 conductance (m s^{-1}). G was obtained through the empirical model of Choudhury et al. (1987), which relates
195 LAI with Rn as:

$$G = 0.079 \cdot [(-0.24 \cdot \text{LAI})] \cdot \text{Rn} \quad (9)$$

The aerodynamic conductance (g_a) was determined by focusing on Monin-Obukhov (Garrat and Hicks, 1973; Loescher et al., 2005) theory, which is based on vertical wind profile. Such modeling was performed daily and the measurement was restricted to neutral conditions of balance in the atmosphere (Allen et al., 1998):

$$g_a = \frac{k^2 \cdot u\,(Z)}{\left\{\ln\left[\frac{(Z - d)}{z_{OM}}\right]\right\}^2} \quad (10)$$

"u" is the wind speed (m s^{-1}) in the Z high (m) (22 m in MOT), d is the zero-plane displacement (m), z_{OM} is surface roughness (m), and k is the von Kármán constant (0.41). Values of z_{0M} and d considered were $(0.1)\cdot h_c$ e $(2/3)\cdot h_c$, respectively; h_c corresponds to the canopy average high, approximately 10.2 m.

There is a relevant difficulty in the canopy conductance parameterization, and several models with empirical approaches that link various parameters that influence the exchange of gases in the stomata have been proposed (Tan et al., 2019). In this study, g_s was obtained by the Penman-Monteith (Eq. 8), using indirect values of λE extracted from the energy balance with residual focus (Fischer et al., 2013; Su, 2002):

$$R_n - G - H = \lambda E \quad (11)$$

This involves the measurement of sensible heat flux H (MJ m^{-2} d^{-1}) using the "Inversing Bulk Transfer Equation" method (based on the Monin-Obukhov similarity theory) (Norman et al., 1995). This method has been applied in different research with micrometeorological approach and remote sensors for the energy flux evaluation (Kalma et al., 2008; Silberstein et al., 2003), being H expressed as:

$$H = c_p \cdot \rho_a \cdot g_a \cdot (T_1 - T_2) \quad (12)$$

T_1 is the air temperature near an aerodynamic surface at canopy height (°C); T_2 is the temperature near the surface (°C). T_1 is the daily average temperature measured on MOT, while T2 is the daily average temperature measured inside the forest at 2.0 m above the ground.

To support the knowledge of the physical controls of evapotranspiration in SSMF, the decoupling coefficient (Ω) and Priestley-Taylor coefficient (α) were calculated. Thus, the evapotranspiration sensitivity to changes in canopy characteristics and local atmosphere conditions can be evaluated through the Ω (McNaughton and Jarvis, 1983):

$$\Omega = \frac{1}{1 + \left[\frac{\gamma}{(\Delta+\gamma)}\right] \cdot \left[\frac{g_a}{g_s}\right]}$$

(13)

According to Ma et al. (2015) and Marques et al. (2020), Ω varies from 0 to 1. Values close to 0 indicate that evapotranspiration is mainly controlled by stomata closure, whereas values close to 1 suggest that energy balance is the limiting factor of evapotranspiration.

Furthermore, to assess if atmospheric demand or soil moisture supply controls evapotranspiration, the α parameter was calculated (Priestley and Taylor, 1972). This is equivalent to the ratio between evapotranspiration and the equilibrium evapotranspiration (E_{eq}), i.e., depends on net radiation and temperature:

$$\alpha = \frac{\lambda E}{\lambda E_{eq}} = \frac{\lambda E}{\left[\frac{(R_n - G)\,\Delta}{(\Delta + \gamma)}\right]}$$

(14)

where λE is the modeled latent heat flux (Eq. 11). Rn, G, Δ and γ were described previously. Values of α < 1 denote ecosystem limitations to the water supply, and α > 1 represents a humid ecosystem, i.e., there are no water limitations, and only the available energy limits evapotranspiration (Marques et al., 2020).

Spatial variability of $ET_{(WB)}$ and precision statistics

The $ET_{(WB)}$ values were mapped in the SSMF, referencing each hydrological year's dry season (from April to September) between 2014 and 2020. The spatial variability was evaluated through geostatistical procedures (ordinary kriging method) and the coefficient of variation (CV) among 31 sampled points of the water balance (Figure 1). The interpolations were performed in ArcGIS (version 10.8) using the exponential semivariogram model, which is widely used for hydrological and soil physical attributes mapping (Junqueira et al., 2017; Kamali et al., 2015; Rodrigues et al., 2020).

A comparative analysis was performed between modeled evapotranspiration (ET) and evapotranspiration (ET_{WB}) obtained from water balance during months that did not present significant precipitation. The slope hypothesis (line 1:1) and its respective coefficient of determination (R^2) were evaluated. The performance of ET to simulate evapotranspiration in the SSMF was evaluated by Root Mean Square Error (RMSE) and Nash-Sutcliffe (NS) coefficient:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\left\{\sum_{i=1}^{N}(P_i - O_i)\right\}^2} \qquad (13)$$

$$\mathrm{NS} = 1 - \frac{\sum_{i=1}^{N}(P_i - O_i)^2}{\sum_{i=1}^{N}(O_i - O_m)^2} \qquad (14)$$

where P_i and O_i are simulated and observed evapotranspiration at time *i*, respectively; N is the number of simulated and observed pair values; and O_m is the observed average values. Values of RMSE close to zero mean a good fit, as well as NS values close to 1.

Results and discussion

Meteorological conditions observed in the studied period

The daily average GP was 3.11 mm d^{-1} during the studied years, showing substantial seasonal variations (Figure 2a). Notably, two GP events exceeding 100 mm d^{-1} were recorded in 2019 (November and December) and one in 2020 (February) in the surroundings of the SSMF. The average air temperature (T) above the forest canopy (Figure 2b) was 21.9°C in the wet season and 18.5°C in the dry season, approximately 0.05ºC lower and 2.51ºC higher than the long-term average, respectively (INMET, 2022).

Within the SSMF, T ranged from 21.9°C in the wet season to 15.8°C in the dry season. Solar radiation (Rs) (Figure 2c) presents daily seasonal variation with higher radiation intensity in the wet season (18.47 MJ m^{-2}) and the lowest in the dry season (15.15 MJ m^{-2}). The average vapor pressure deficit (VPD) was 0.58 kPa and 0.54 kPa in the dry and wet seasons, respectively (Figure 2d).

Figure 2e presents the relative extractable water (REW) at five soil depths (0.1 m, 0.2 m, 0.3 m, 0.4 m, and 1.0 m). During the wet season, frequent rainfall maintains REW close to 1 across all depths, fulfilling the field capacity. In contrast, during the dry season, the forest's water uptake causes REW to decrease gradually, especially in the 0.1, 0.2, and 0.3 m depths, where average values decrease to 0.24, 0.44, and 0.48, respectively. However, at the 1.0 m depth, REW remains approximately 0.7 to 0.8. This indicates that trees' root systems slightly uptake water from this depth, focusing more on the shallow layers even in the dry season. Thus, the water balance in the dry season can be accurately assessed by considering a soil layer confined to 1.0 m without losing accuracy in ET estimates.

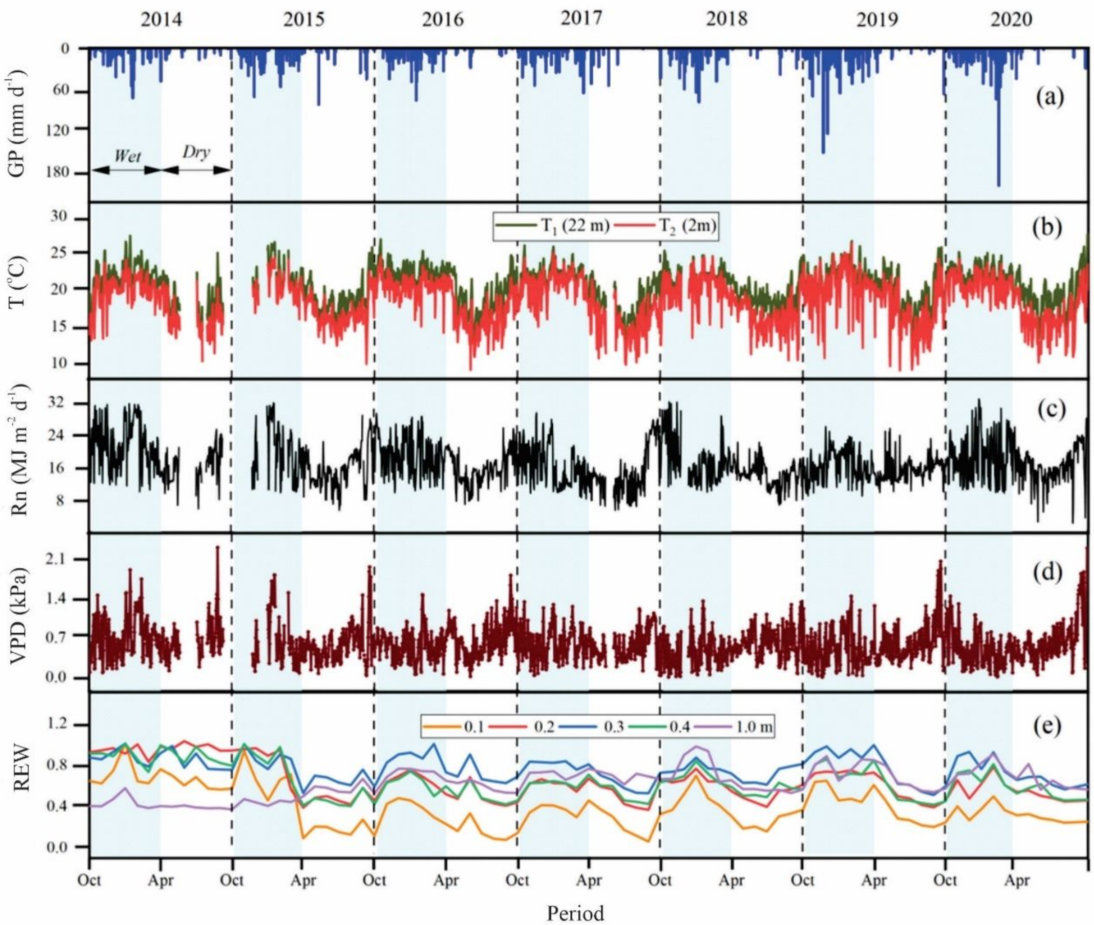
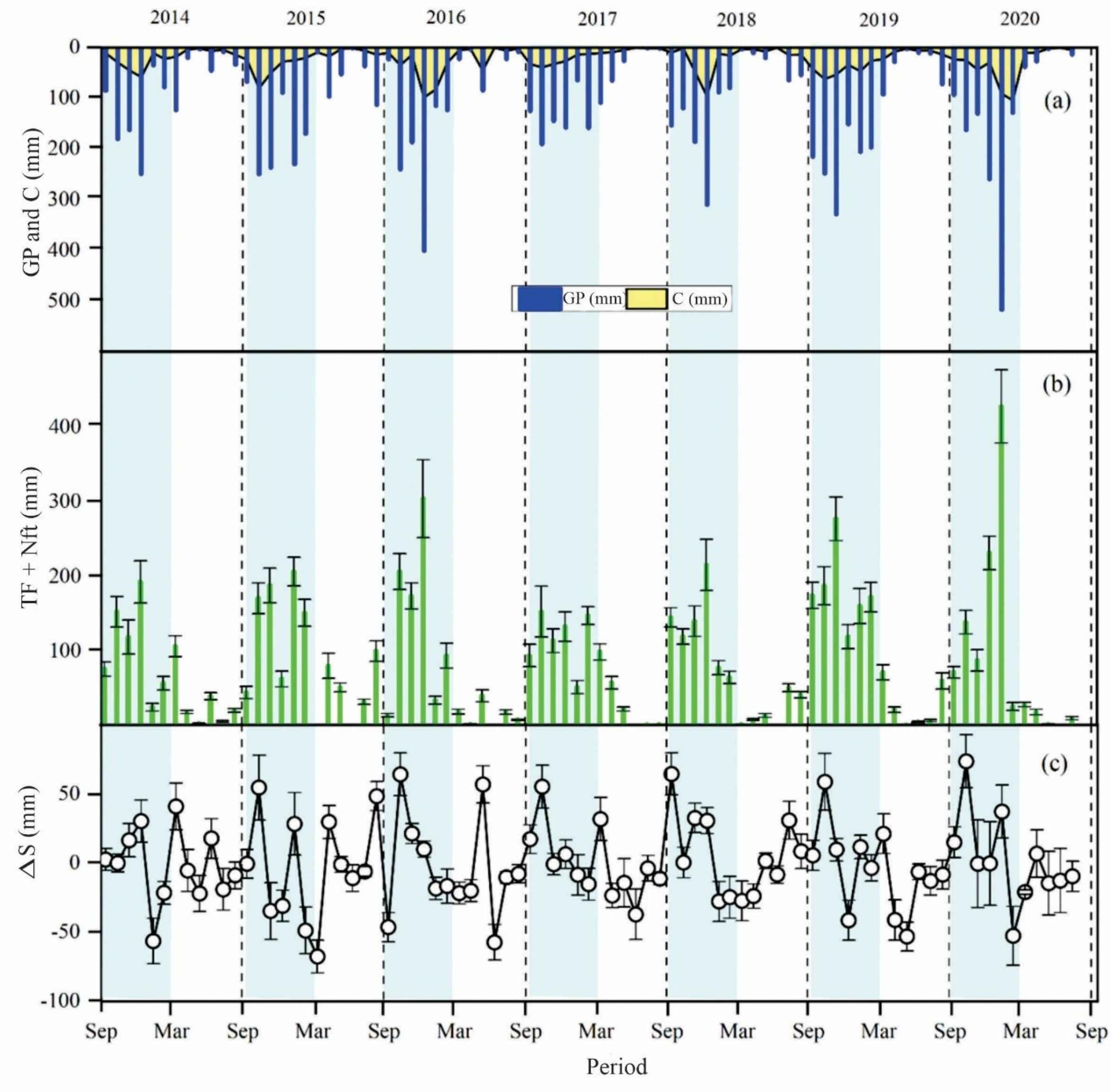


Figure 2. Average daily (a) GP, (b) air temperature (T), (c) global solar radiation (R_s), (d) vapor pressure deficit (VPD), and (e) relative extractable water (REW) in the 0–1.0m soil layer. Shaded columns represent wet seasons.

Precipitation partitioning and soil water storage in the studied period

The GP values (Figure 3a) for each hydrological year were as follows: 1024 mm (2013-2014), 1350.1 mm (2014-2015), 1229.5 mm (2015-2016), 1043.5 mm (2016-2017), 1087.4 mm (2017-2018), 1561.5 mm (2018-2019), and 1399.8 mm (2019-2020). GP was notably below the average of 1461.8 mm for the years 2013-2014, 2016-2017, and 2017-2018, resulting in significantly drier conditions. However, the GP values for the last two hydrological years is near the average, which were insufficient to fully replenish the soil water content and restore the hydrological processes in the SSMF. The daily average values for the hydrological years are in

(Supplementary).

296 

297 Figure 3. Monthly gross precipitation (GP) and canopy rainfall interception (C) (a), monthly effective (or
298 net) precipitation (NP = TF + St), with respective standard deviations (vertical bars) (b), monthly soil water
299 storage variation in the 0-1.0 m control layer (ΔS), with respective standard deviations (vertical bars) (c).
300 Shaded columns represent wet seasons.

301 Seasonal precipitation patterns in SSMF roughly follow the regional precipitation trends. However,
302 over the past decade, the southeastern region of Brazil has experienced a significant decrease in
303 precipitation compared to the long-term average. The hydrological years of 2013-2014, 2016-2017, and

2017-2018 were characterized by severe to moderate meteorological droughts, particularly during the wet season (Guauque-Mellado et al., 2021). Notably, 2017 saw the lowest annual precipitation in decades, approximately 30% of the long-term average, provoking a change in the temporal distribution of the rainfall in the wet season and a severe reduction in soil water storage.

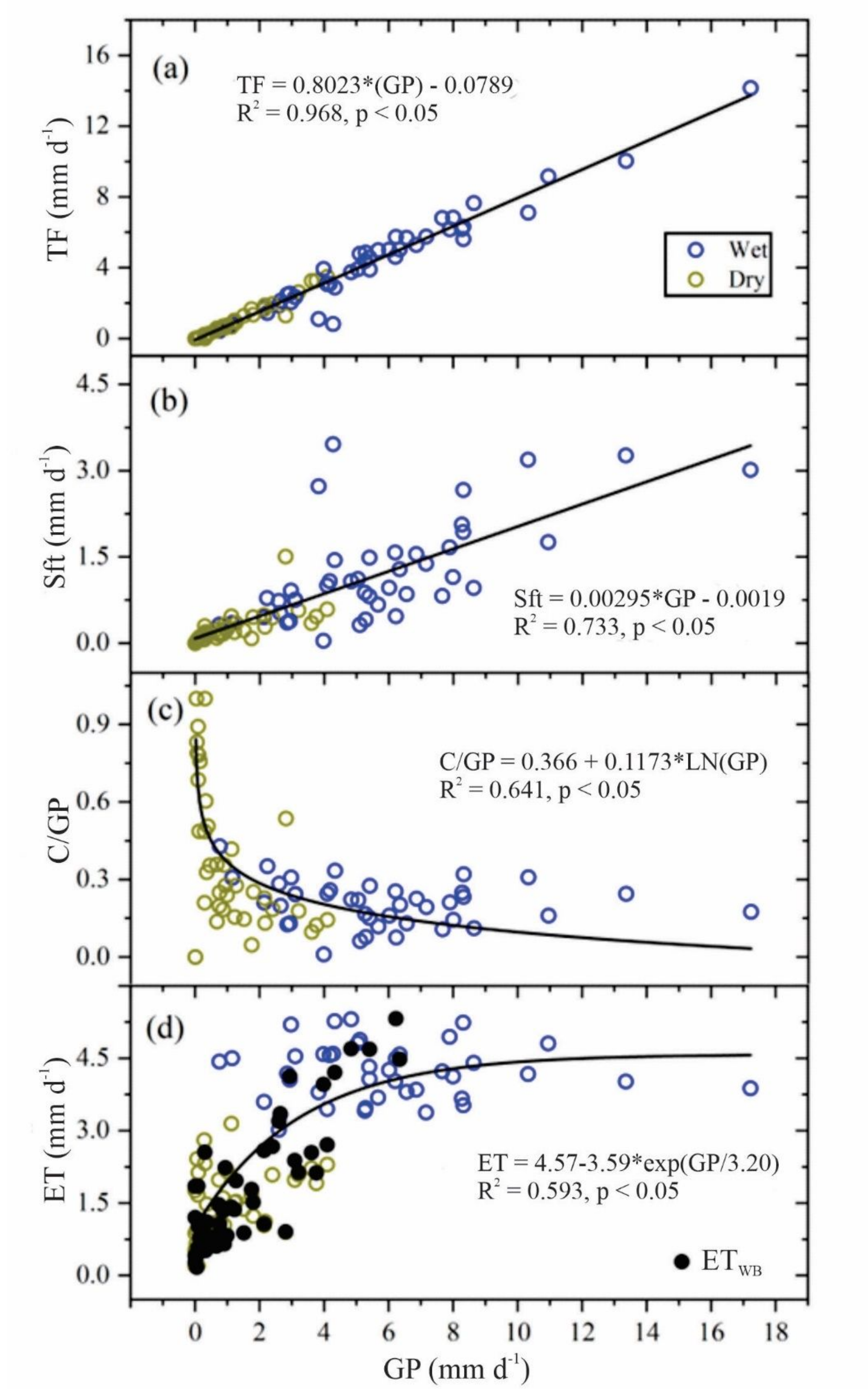
C had a proportional variation with rainfall amount across the studied years, averaging 19.13%. In the hydrological year of 2019-2020, C increased to 25.3% due to a more significant rainfall amount. The net precipitation represented 81.22% of GP, of which TF accounted for 99.9%, ranging from 765.4 mm (2015-2016) to 1239.2 mm (2018-2019). The average TF in the wet season was 670 mm (82.06% of GP); in the dry season, it was 149.6 mm (75.21% of GP). The annual average Sf was 2.6 mm during wet and 0.4 mm during dry seasons, corresponding to 0.1% of GP in the studied period.

The monthly soil water storage variation (ΔS) stood out with a marked predominance of positive values (i.e., increased soil water storage) in the wet season (Figure 3c), with an average of 28 mm. However, at the end of the 2013-2014 and 2014-2015 hydrological years, ΔS presented, respectively, -28.5 mm and -32.2 mm. In the dry seasons, the lack of rainfall presented a higher recession of soil water storage by water consumption in the ecosystem, averaging -41.4 mm. Interestingly, ΔS response to GP was more sensitive during the dry season, highlighting positive variations close to (e.g., 2015 and 2018) or even greater than (e.g., 2016) those observed in the wet season. This is likely due to the drier soil and lower evapotranspiration, which translates into a greater reservoir (empty porous) and the latent response of the forest to the new water availability.

The seasonality of ΔS (0-1.0m) followed variations in GP, corresponding with minor changes in rainfall patterns. Consequently, drought conditions can lead to significant soil water deficits, even during the wet season, due to short dry spells common in the region (Junqueira Junior et al., 2019). For instance, January 2016 and February 2020 exhibited ΔS values comparable to those in drier months. Junqueira et al. (2017) studied the temporal stability of soil moisture in the SSMF from 2012 to 2015. They observed that forest influences soil moisture stability at shallower depths (up to 0.40 m), while soil attributes, such as saturated hydraulic conductivity, govern moisture behavior in deeper depths. Similarly, Mello et al. (2019) observed comparable results in another Atlantic Forest remnant with shallow soil (Inceptisols). These findings underscore the crucial role of Atlantic Forests in shaping the temporal and spatial behavior of soil moisture, thereby affecting water balance outcomes.

TF and Sf showed a linear relationship with GP (Figure 4a, b), following the seasonal pattern of GP. On the other hand, the relationship between C/GP and GP had a significant log relationship because more drainage pathways are formed as the canopy saturates (Rodrigues et al., 2022). As GP increases, canopy interception becomes less important, and more water is transformed into TF and SF. This impacts the rainfall redistribution within the forest and the spatial dynamics of soil moisture (Fischer-Bedtke et al., 2023). The variabilities observed in C/GP are due to the biodiversity and structure of the ecosystem, leading to complex canopy properties such as differences in the foliage size, roughness of trunks, and funneling effect, among other vegetative aspects (Terra et al., 2018). In the SSMF, C/GP had an average value in the studied period of 19.3%, a value close to that obtained in an Atlantic Rainforest at the Mantiqueira Mountain Range in southeastern Brazil (20.9%, Mello et al., 2019). Salemi et al. (2013) found up to 33% of this ratio in a neotropical forest at Serra do Mar Mountain Range, also in southeastern Brazil. This difference can be attributed to the canopy aspects, as in Serra do Mar, an ombrophilous forest was monitored in lower elevations, meaning a canopy with large leaves and a closer canopy over the year (Mello et al., 2019). Moreover, Iida et al. (2020) observed values of 5.3% in a deciduous dry rainforest in Kratie, Cambodia.

A significant exponential relationship between evapotranspiration and GP is observed in Figure 4d, highlighting the transition from a water-limited environment to an energy-limited environment. This process could be detected when analyzing the differences in the evapotranspiration of the wet (blue dots) and dry periods (green dots) (Figure 4d). Evapotranspiration increases with GP due to higher REW, which enhances stomata opening and gas exchange during photosynthesis. A stabilization is reached for significant amounts of precipitation from which REW is no longer the limiting factor for evapotranspiration.



353

354 Figure 4. Relationships between average observed gross rainfall (GP): (a) throughfall (TF); (b) stemflow;

355 (c) C/GP ratio (Stf); and (d) actual evapotranspiration (ET).

356

Evapotranspiration modeling

Figure 5 presents the daily average evapotranspiration from the water balance analysis performed during the dry season (i.e., April-September, from 2014 to 2020). Evapotranspiration varied between 0.90- and 1.75-mm d^{-1}, with the hydrological years 2017-2018 and 2014-2015 recording the minimum and maximum values, respectively. The coefficient of variation (CV) varied from 5% to 14.17%, with the lowest values occurring in the driest years. This variability in evapotranspiration can be attributed to the spatial distribution of water inputs in the soil, such as canopy heterogeneity, and the distribution of soil physical properties, like soil moisture. Overall, ET variability in SSMF is low, primarily due to the uniform physical properties of the Red Latosol both spatially and with depth, which also results in low variability of the REW during the dry season.

Evapotranspiration estimates in this study aligned with findings from prior research in the Atlantic Forest ecosystem. For example, Pereira et al. (2010), Salemi et al. (2013), and Mello et al. (2019) reported daily average values of 3.9, 3.56, and 3.26 mm d^{-1}, respectively. In this study, daily evapotranspiration varies from 4.96 mm d^{-1} in the wet season to 1.55 mm d^{-1} in the dry season, aligning with ET estimated in the Atlantic Forest biome.

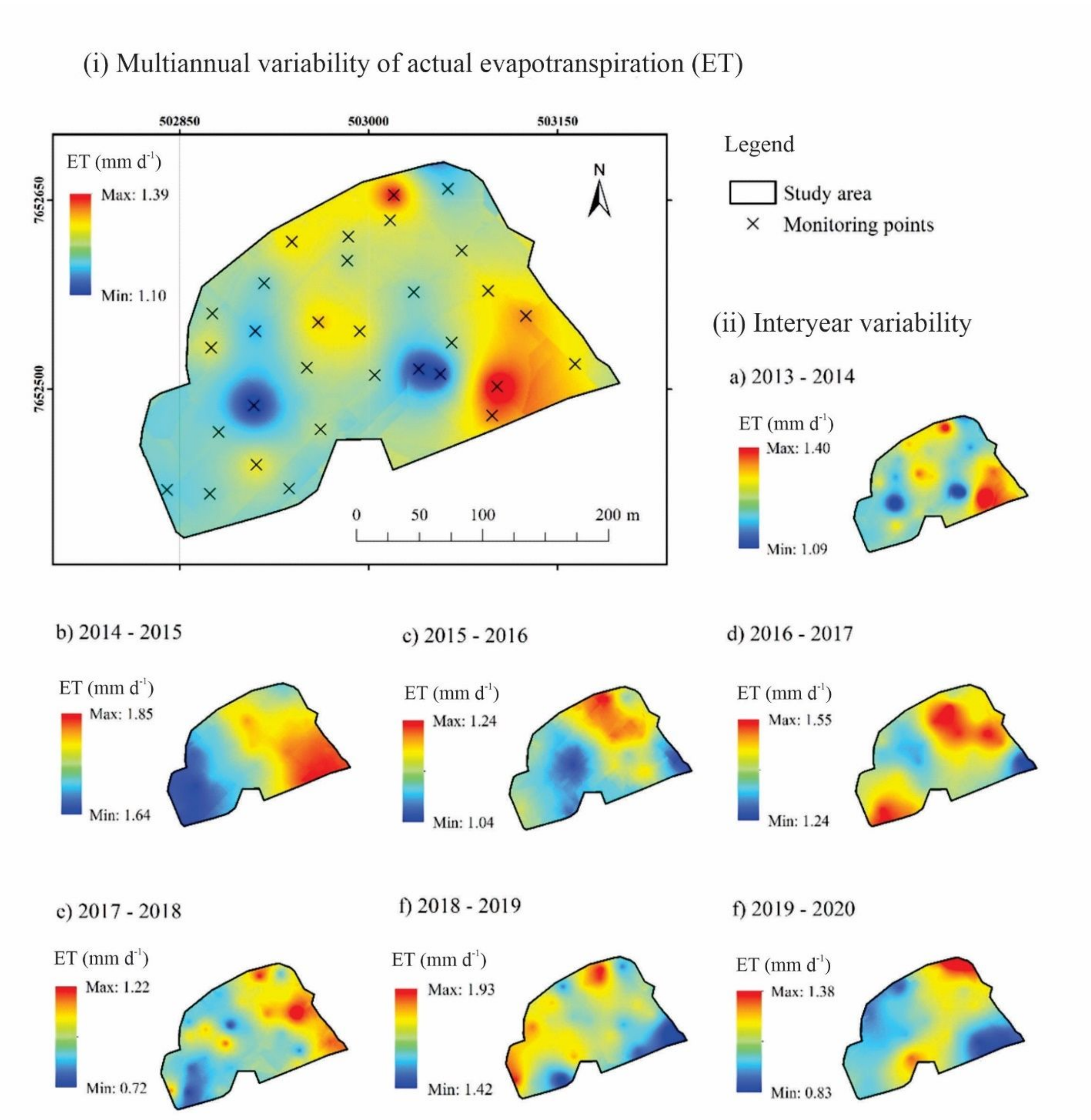


Figure 5. Spatial distribution of daily average evapotranspiration in the dry period: (i) multiannual variability (average from 2014 to 2020); (ii) interannual variability.

The temporal behavior of the modeled daily evapotranspiration, energy fluxes, Priestley-Taylor coefficient (α), canopy conductance (g_s), and the decoupling factor (Ω) are displayed in Figure 6. The Rn values showed clear seasonality, with the highest values in the wet season (16.21 MJ m^{-2} d^{-1} on average) and the lowest in the dry season (9.21 MJ m^{-2} d^{-1} on average) (Figure 6a). The heat flux in the soil is near zero, corresponding on average to 0.43 MJ m^{-2} d^{-1} in wet seasons and 0.18 MJ m^{-2} d^{-1} in dry seasons. H presents a higher seasonality fluctuation, with a daily average value of 6.16 MJ m^{-2} d^{-1} in the dry season, whereas in the wet season, 5.37 MJ m^{-2} d^{-1}. On the other hand, higher values of λE (Figure 6b) occurred in the wet season, with an average of 10.28 MJ m^{-2} d^{-1} (equivalent to 4.20 mm d^{-1}) and 3.11 MJ m^{-2} d^{-1} (equivalent to 1.27 mm d^{-1}) for dry season.

Regarding Rn, λE corresponded to 60% of it in the study period. The Priestley-Taylor coefficient (α) presented values > 1 in the wet season, indicating that evapotranspiration is mostly controlled by atmospheric demand (Figure 6c) in this period. An average value of 0.51 in the dry seasons means that soil moisture also controls evapotranspiration, but atmospheric demand still plays a significant role in this period. In addition, g_s (Figure 6d) presented the same seasonal pattern as evapotranspiration, with an average in the wet season of 3.47 mm s^{-1} and, in the dry season, 0.85 mm s^{-1}. The seasonal pattern of Ω (Figure 6e) was similar to g_s. The low values indicate a substantial control of evapotranspiration through g_s in dry (0.10) and wet (0.35) seasons. Table A2 details the daily average values for these components throughout the seven hydrological years studied.

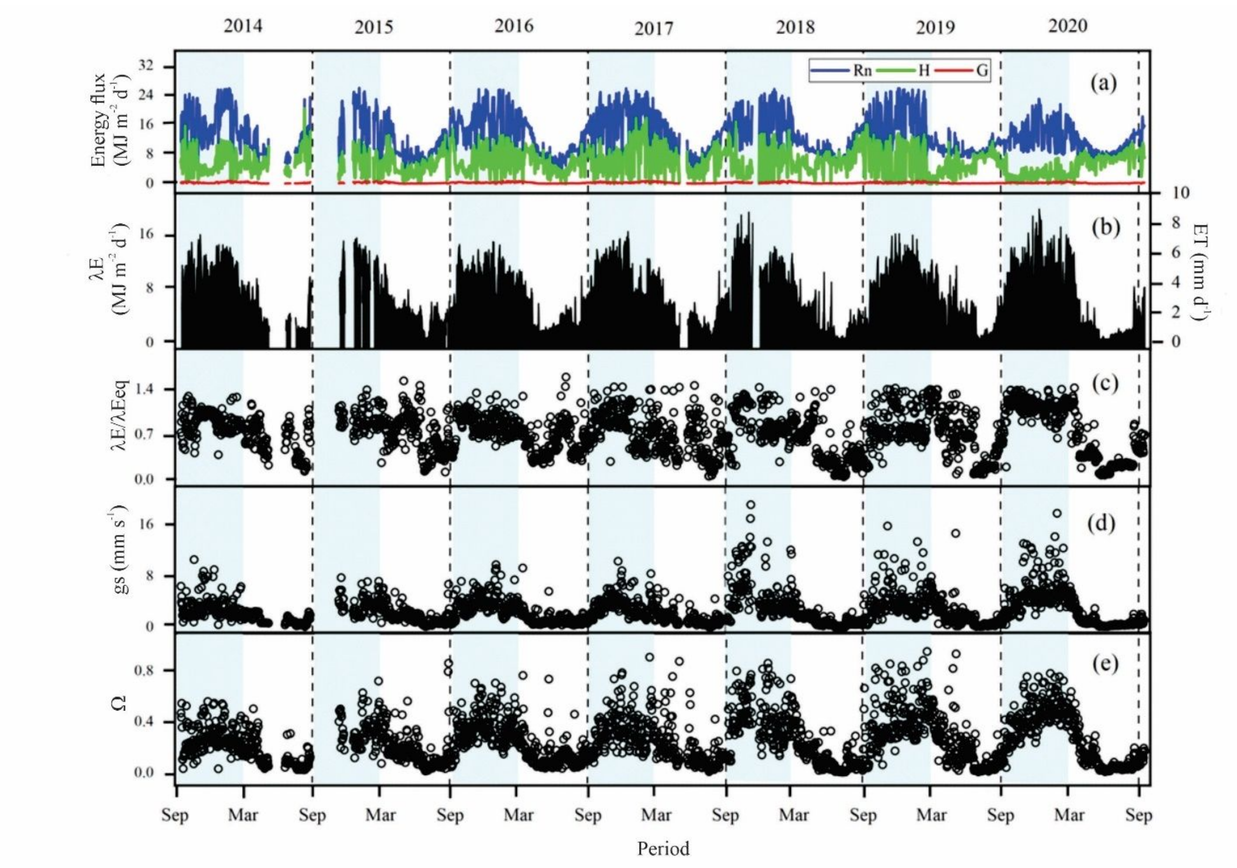


Figure 6. Annual modeled data of (a) daily energy flux: net radiation, sensible heat flux and soil heat flux (Rn, H and G; left scale); (b) latent heat flux (λE, left scale) and equivalent evapotranspiration in millimeters (ET, right scale); (c) daily coefficient of Priestley-Taylor ($\alpha = \lambda E/\lambda E_{eq}$, left scale); (d) daily canopy conductance (g_s, left scale); and (e) daily decoupling factor (Ω, left scale). Shaded columns represent periods of wet seasons.

The meteorological data and estimated canopy conductance support the evapotranspiration modeling approach for SSMF (i.e., a semi-deciduous forest). The Penman-Monteith method could adequately estimate the actual evapotranspiration of the SSMF. Figure 7 highlights the relationship between ET_{PM} (Penman-Monteith evapotranspiration) and ET_{WB} (water balance evapotranspiration). The regression line is not significantly different from line 1:1, indicating a good performance of the modeled evapotranspiration. Moreover, the value of RMSE was 0.41 mm d^{-1}, and the NS coefficient was 0.86, suggesting that the modeled evapotranspiration was close to those from water balance in the dry season.

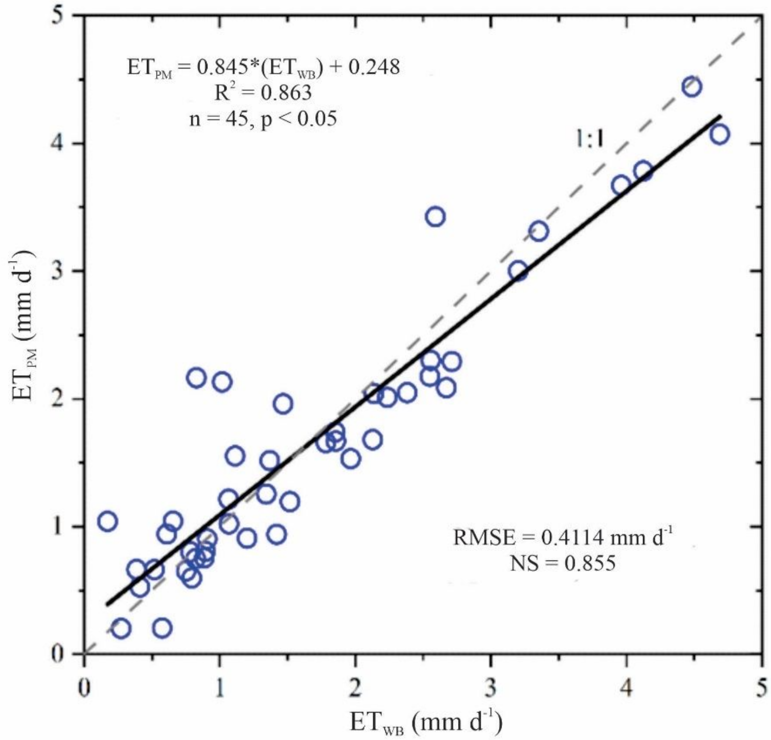


Figure 7. Relationship between modeled evapotranspiration (ET_{PM}) and evapotranspiration from water balance (ET_{WB}) and precision statistics (e.g., RMSE and NS).

The implementation of the evapotranspiration model accurately simulated observed values from water balance analysis, exhibiting a deviation of 13.7%. This error margin aligns with findings from Mello et al. (2019), Muñoz-Villers et al. (2012), and Pereira et al. (2010). The model facilitated an investigation into the seasonal variations in evapotranspiration observed in the forest over seven hydrological years, predominantly driven by fluctuations in radiation and soil moisture levels. Specifically, radiation accounted for 60% of the evapotranspiration responses, a finding consistent with other studies conducted in Neotropical forests across Brazil and Mexico, particularly in semi-deciduous forest sites (Cabral et al., 2015; Rocha et al., 2004; Muñoz-Villers et al., 2012).

The gas exchange rates observed during the study aligned with those reported for secondary tropical forests, exhibiting average daily values ranging from 0.9 mm s^{-1} to 12.7 mm s^{-1} (Cabral et al., 2015; Rocha et al., 2004; Fraga et al., 2015; Marques et al., 2020). Tan et al. (2019) noted that in tropical forests, variations in precipitation substantially impact gas exchange, i.e., photosynthesis and transpiration, particularly during the dry season, as a means for the forest to mitigate water and energy loss. This phenomenon was evident in our study, which showed a marked decline in canopy conductance from March to September, coinciding with the dry season. Furthermore, we observed that evapotranspiration rates increase exponentially in response to changes in vapor pressure deficit (VPD), particularly when VPD exceeds 0.7 kPa (**Error! Reference source not found.**1S). However, the gas exchange decreases with VPD, showing a stomatal regulation due to the environmental conditions (i.e., increased atmospheric demand). This forest mechanism regulates the water transfer into the atmosphere, controlling the root water uptake and soil water depletion. Corroborating our findings, Cabral et al. (2015) and Igarashi et al. (2015) documented peak evapotranspiration rates preceding minimal VPD levels during rainy periods, a result of enhanced stomatal opening driven by the water vapor transport gradient.

In terms of biophysical evapotranspiration controls, the seasonal variations of Ω and α (λE / λE_{eq}) were more pronounced in the rainy period (Ma et al., 2015; Tan et al., 2019). The results suggest that in SSMF, stomatal regulation predominantly governed evapotranspiration. The distribution of α showed a greater atmospheric demand in wet periods than in dry periods, influencing stomatal response to water availability. This relationship between α and relative extractable water (REW) across five soil depths was clearly demonstrated in **Error! Reference source not found.**2S. Notably, water depletion occurred more rapidly in shallower layers (0-0.2 m). Maseyk et al. (2008), Aguilos et al. (2019), and Jiao et al. (2019) mentioned that ET is strongly constrained by soil moisture, particularly during water deficits that lead to decreased canopy conductance to mitigate water stress from soil moisture depletion. This mechanism is crucial in ecosystems like the Atlantic Forest, which experiences seasonal climate variation and periods of reduced precipitation. Therefore, more extended drought periods, potentially due to climate change, can significantly impact water availability in the soil, thereby affecting evapotranspiration and photosynthesis.

The water balance of the SSMF

Figure 8 presents the components of the monthly water balance in SSMF. During the studied period, the maximum values of GP and ET were observed in 2018-2019, with values of 1561.5 and 1055.8 mm, respectively. Notably, monthly ET followed the seasonal pattern of GP, as the summer is the wet season, i.e., the atmospheric demand and REW are higher. The monthly ΔS was predominantly positive in wet periods and negative in dry periods (Figure 8c). Annually, it stands out with positive values in 2016-2017, 2017-2018, and 2019-2020. In the transition between wet and dry periods, ΔS was marked by negative values as evapotranspiration was kept high in response to soil water and energy availability.

Percolation (P) is predominantly positive on an annual scale, with an average value of 323.76 mm in the study period. In drier years (2013-2014, 2016-2017, and 2017-2018), annual P decreased significantly (Figure8d). ET showed marked participation in the drier years, which affected the percolation due to higher atmospheric demand and lower than average GP.

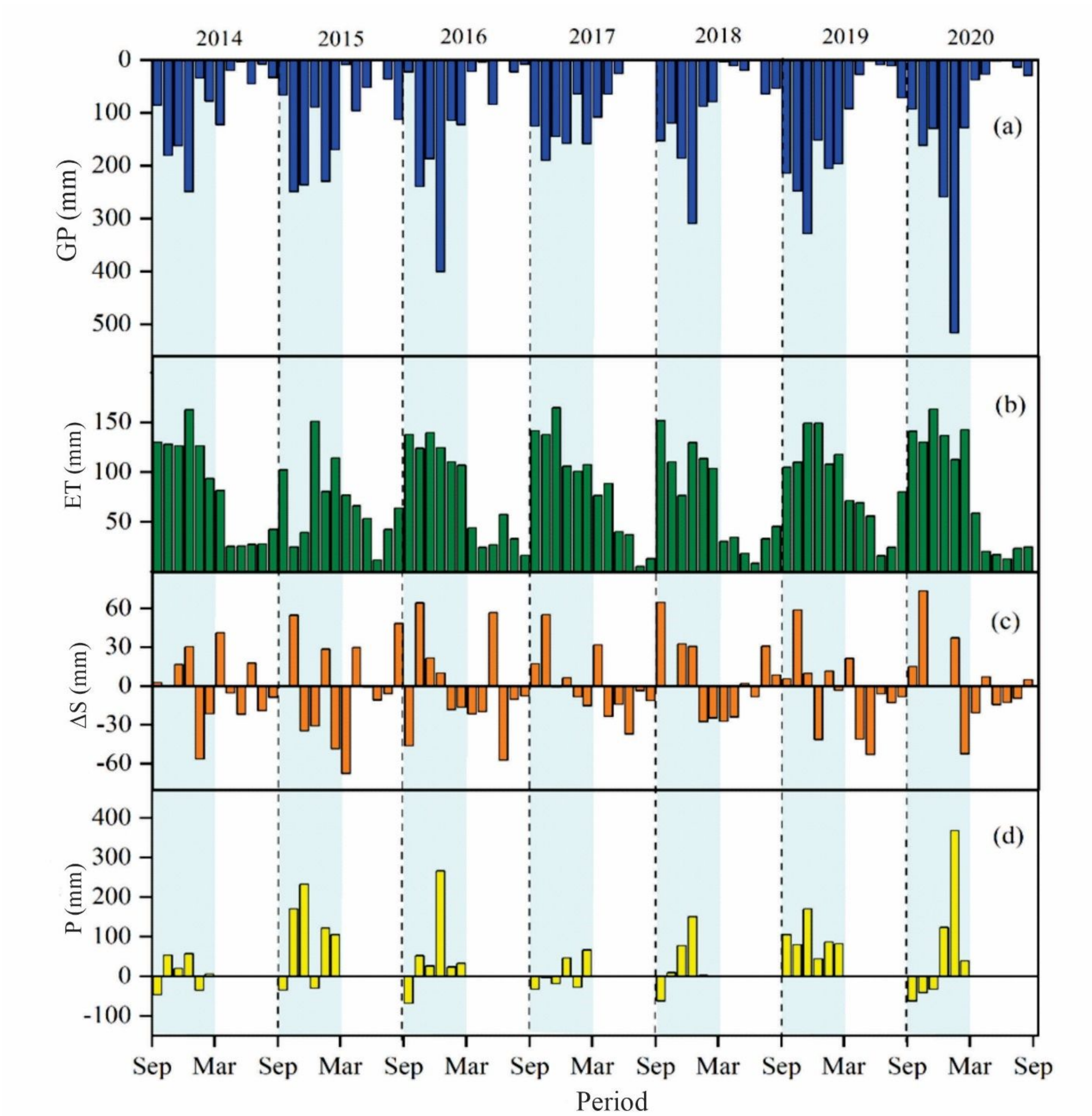


Figure 8. Monthly water balance components: (a) Gross precipitation (GP); (b) evapotranspiration (ET); (c) water storage variation (ΔS) in the root zone; and (d) the water percolation (P) for the seven hydrological studied years (2014 to 2020).

The analysis of water balance elements revealed significant variances across the seven hydrological years studied. Annually, the capillary effect was shown to be negligible, even under severe drought conditions (Figure 8d). However, in months where ET exceeds GP, P becomes negative. This occurs mainly in months after the dry period (September and October), where evapotranspiration tends to increase due to changes in radiation. Therefore, small changes in rainfall can reduce the water percolation in this ecosystem. In addition, our results, aligned with those of other studies (e.g., Li et al., 2010; Negrón Juárez et al., 2007), suggest that deep soil water storage in the previous wet season generally sustains forest evapotranspiration during subsequent dry periods, thereby mitigating the effects of rainfall deficits. The temporal behavior of REW at 1.0 m in depth has little variability, suggesting that the water stored in this layer suffices to support the ecosystem in the dry period. The highest P values followed those of GP, primarily in the middle of the wet season. In these moments, REW was high (due to previous rains), and evapotranspiration was driven mainly by energy availability. Therefore, excess water ($\theta > \theta_{FC}$) was transported throughout the profile by activation/optimization of preferential pathways, increasing P. This suggests efficient macropore drainage as saturation was never reached with increased GP (Jiang et al., 2023), driving water directly to deeper layers through preferential flow.

The Atlantic Forest, including the specific areas of SSMF, is critical in regulating water use during drought conditions. No significant water deficit was observed in the SSMF over the study period, even during relatively dry years. However, drought conditions did impact the SSMF's hydrological processes by reducing canopy interception, soil moisture, and percolation rates (Guauque-Mellado et al., 2021). On the other hand, the variability in evapotranspiration observed did not align with the growth patterns typically seen in secondary tropical forests, a phenomenon reported in other research (e.g., Kume et al., 2011; Von Randow et al., 2020). This disparity may be attributed to several factors: (i) the edge effect, which hinders forest growth and diversification; (ii) climatic variability; and (iii) the trees' physiological strategies to battle with water deficit.

We analyzed the water balance, considering that up to 1.0 m depth of the root system supplies most of the water consumed for this forest ecosystem. However, some species may seek water at deeper layers than 1.0 m, as the deep roots in tropical forests can have a fundamental role in ET (e.g., Rocha et al., 2004;

Kume et al., 2011). In this way, further studies should progress in the hydrological field of Atlantic forests by analyzing the water balance, considering deeper layers, preferential flows existence, and the role of the roots in this process.

This study presents some limitations and uncertainties that need to be highlighted: (i) ∂S was obtained in a monthly step, which precludes the analysis of the temporal variability in shorter scales; (ii) the soil layer up to 1.0 m depth, which is a source of uncertainty for P estimates as tropical species can reach deeper layers (Broedel et al., 2017); (iv) measurement of ET using a more precise approach, such as eddy covariance and sap flow. Despite these shortcomings, the analysis highlights the importance of evapotranspiration in the SSMF, conducted in a forest site in Brazil for the first time throughout seven consecutive hydrological years. These results could improve the understanding of tropical forests under climate change and support decision-makers on better management practices for Atlantic Forest biomes.

Conclusions

The study delineates evapotranspiration as a pivotal component of the water balance in Atlantic Forest biomes, exhibiting spatial variability and critical linkage with factors such as soil moisture, vapor pressure deficit, air temperature, and atmospheric pressure. This process is essential in regulating hydrological mechanisms, particularly soil water storage during periods marked by water scarcity. The primary insights from the investigation are outlined as follows:

a) Net precipitation accounted for 81% of GP, comprising 99.9% throughfall and 0.1% streamflow. Canopy interception was significant in the forest and corresponded to 19% of GP over the studied period; importantly, during the driest hydrological years, evapotranspiration constituted more than 90% of gross precipitation.

b) Throughout the studied period, most hydrological years ended with negative soil water storage changes at 1.0 m in depth, except for hydrological years from 2018 to 2020.

c) The study observed minimal interannual spatial variations in evapotranspiration at the SSMF (CV=4.72%). Evapotranspiration estimates were reliable across different scales, enhancing the understanding of water balance within the forest ecosystem.

d) Evapotranspiration accounted for 60% of Rn. During the dry seasons, it was primarily governed by soil water availability and stomatal regulation, whereas in wet seasons, the available energy and atmospheric conditions largely influenced its behavior.

e) Water percolation below the root zone showed no significant capillary rise, mainly because of the depth of the Oxissols. It varied from 3% to 49% of GP on a monthly scale, with notable reductions in drier years.

References

Aguilos, M., Stahl, C., Burban, B., Hérault, B., Courtois, E., Coste, S., Wagner, F., Ziegler, C., Takagi, K., Bonal, D. (2019). Interannual and seasonal variations in ecosystem transpiration and water use efficiency in a tropical rainforest. *Forests,* 10, 14; doi:10.3390/f10010014.

Allen, R.G., Pereira, L.S., Raes, D., Smith, M. Crop evapotranspiration guidelines for computing crop water requirements. Rome: FAO, 1998.

Borges, C. K. et al. (2020) Seasonal variation of surface radiation and energy balances over two contrasting areas of the seasonally dry tropical forest (Caatinga) in the Brazilian semi-arid. *Environmental Monitoring and Assessment*, 192, 8.

Broedel, E. et al. (2017) Deep soil water dynamics in an undisturbed primary forest in central Amazonia: Differences between normal years and the 2005 drought. *Hydrological Processes*, 31, 9, 1749–1759.

Cabral, O.M.R., Rocha, H.R., Gash, J.H., Freitas, H.C., Ligo, M.A.V. (2015) Water and energy fluxes from a woodland savanna (cerrado) in southeast Brazil. *Journal of Hydrology: Regional Studies*, 4, 22–40.

534 Choudhury, B. J., Idso, S. B., Reginato, R. J. (1987) Analysis of an empirical model for soil heat flux under a growing wheat crop for estimating evaporation by an infrared-temperature based energy balance equation. *Agricultural and Forest Meteorology,* 39, 4, 283–297.

537 Costa, M.H., Biajoli, M.C., Sanches, L., Malhado, A.C.M., Hutyra, L.R., Rocha, H.R., Aguiar, R.G., Araújo, A.C. (2010) Atmospheric versus vegetation controls of Amazonian tropical rain forest evapotranspiration: Are the wet and seasonally dry rain forests any different? *Journal of Geophysical Research: Biogeosciences,* 115, 4, 1–9.

541 Ershadi, A., McCabe, M.F., Evans, J.P., Wood, E.F. (2015) Impact of model structure and parameterization on Penman-Monteith type evaporation models. *Journal of Hydrology*, 525, 521–535.

543 Fischer-Bedtke, C., Metzger, J. C., Demir, G., Wutzler, T., and Hildebrandt, A. (2023) Throughfall spatial patterns translate into spatial patterns of soil moisture dynamics – empirical evidence, *Hydrol. Earth Syst. Sci.*, 27, 2899–2918.

546 Fischer, M., Trnka, M., Kucera, J., et al. (2013) Evapotranspiration of a high-density poplar stand in comparison with a reference grass cover in the Czech-Moravian Highlands. *Agricultural and Forest Meteorology,* 181, 43–60.

549 Fraga, C.I.M., Sanches. L., Pinto Junior, O.B., Curado, L.F.A., Gaio, D.C. (2015) Condutância do dossel, condutância aerodinâmica e fator de desacoplamento em floresta de Vochysia divergens Pohl (Vochysiaceae) no Pantanal Brasileiro. *Revista Brasileira de Meteorologia*, 30, 3, 275–284.

552 Freitas, W.K., Magalhães, L., Santana, C.A.A., Pereira Junior, E.R., Souza, L.C.M., Toledo, R.A.B., Garção, B. (2020) Tree composition of urban public squares located in the Atlantic Forest of Brazil: A systematic review. *Urban Forestry and Urban Greening*, 48, 86, 126555.

555 Garratt, J., Hicks, B. (1973) Momentum, heat and water vapour transfer to and from natural and artificial surfaces. *Quarterly Journal of the Royal Meteorological Society*, 99, 422, 680-687.

Ghimire, C. P., Lubczynski, M.W., Bruijnzeel, L.A., Chavarro-Rincon, D. (2014) Transpiration and canopy conductance of two contrasting forest types in the Lesser Himalaya of Central Nepal. *Agricultural and Forest Meteorology*, 197, 76–90.

Guauque-Mellado, D., Rodrigues, A.F., Terra, M., Mantovani, V., Yanagi, S., Diotto, A., Mello, C. (2022). Evapotranspiration under Drought Conditions: The Case Study of a Seasonally Dry Atlantic Forest. *Forests,* 13, 871.

Igarashi, Y., Kumagai, T., Yoshifuji, N., Sato, T., Tanaka, K., Suzuki, M., Tantasirin, C. (2015) Environmental control of canopy stomatal conductance in a tropical deciduous forest in northern Thailand. *Agricultural and Forest Meteorology*, 202, 1–10, 2015.

Iida, S., Shimizu, T., Tamai, K., et al. (2020) Evapotranspiration from the understory of a tropical dry deciduous forest in Cambodia. *Agricultural and Forest Meteorology*, 295, 108170.

Jiang, Y., Zhang, Y., Fan, B., Wen, J., Liu, H., Mello, C.R., Cui, J., Yuan, C., Guo, Li. (2023) Preferential flows influences the temporal stability of soil moisture in a headwater catchment. *Geoderma*, 437, 116590.

Jiao, L., Lu, N., Fang, W., Li, Z., Wang, J., Jin, Z. (2019). Determining the independent impact of soil water on forest transpiration: A case study of a black locust plantation in the Loess Plateau, China. *Journal of Hydrology*, 572, 671–681.

Jones, H., Black, A., Jassal, R.S., Nesic, Z., Grant, N.J., Bhatti, J., Sidders, D. (2017). Water balance, surface conductance and water use efficiency of two young hybrid-poplar plantations in Canada's aspen parkland. *Agricultural and Forest Meteorology*, 246, 256–271.

Junqueira, J.A., Mello, C.R., Owens, P.R., Mello, J.M., Curi, N., Alves, G.J. (2017). Time-stability of soil water content (SWC) in an Atlantic Forest - Latosol site. *Geoderma*, 288, 64–78.

Junqueira, J. A., Mello, C.R., Mello, J.M., Scolforo, H.F., Beskow, S., McCarter, J. (2019) Rainfall partitioning measurement and rainfall interception modelling in a tropical semi-deciduous Atlantic forest remnant. *Agricultural and Forest Meteorology*, 275, 170–183.

Kalma, J.D., McVicar, T.R., McCabe, M.F. (2008) Estimating land surface evaporation: A review of methods using remotely sensed surface temperature data. *Surveys in Geophysics*, 29, 4–5, 421–469.

Kamali, M.I., Nazari, R., Faridhosseini, A., Ansari, H., Eslamian, S. (2015) The Determination of Reference Evapotranspiration for Spatial Distribution Mapping Using Geostatistics. *Water Resources Management,* 29, 11, 3929–3940.

Kelliher, F.M., Leuning, R., Raupach, M.R., Schulze, E.D. (1995). Maximum conductances for evaporation from global vegetation types. *Agricultural and Forest Meteorology*, 73, 1–2, 1–16.

Kume, T., Tanaka, N., Komatsu, H., Yoshifuji, N., Saitoh, T.M., Suzuki, M., Kumagai, T. (2011). Ten-year evapotranspiration estimates in a Bornean tropical rainforest. *Agricultural and Forest Meteorology*, 151, 9, 1183–1192.

Li, Z., Zhang, Y., Wang, S., Yuan, G., Yang, Y., Cao, M. (2010) Evapotranspiration of a tropical rain forest in Xishuangbanna, southwest China. *Hydrological Processes*, 24, 17, 2405–2416.

Loescher, H. W., Gholz, H.L., Jacobs, J.M., Oberbauer, S.F. (2005). Energy dynamics and modeled evapotranspiration from a wet tropical forest in Costa Rica. *Journal of Hydrology*, 315, 1–4, 274–294.

Ma, N., Zhang, Y., Guo, Y., Gao, H., Zhang, H., Wang, Y. (2015). Environmental and biophysical controls on the evapotranspiration over the highest alpine steppe. *Journal of Hydrology*, 529, 980–992.

Macedo, T. M., Costa, W.S., Brandes, A.F.N., Valladares, F., Barros, C. (2021) Diversity of growth responses to recent droughts reveals the capacity of Atlantic Forest trees to cope well with current climatic variability. *Forest Ecology and Management,* 480, 118656.

600 Markewitz, D., Devine, S., Davidson, E.A., et al. (2010) Soil moisture depletion under simulated drought
601 in the Amazon: Impacts on deep root uptake. *New Phytologist,* 187, 3, 592–607.

602 Marques, T.V., Mendes, K., Mutti, P.R. et al. (2020) Environmental and biophysical controls of
603 evapotranspiration from Seasonally Dry Tropical Forests (Caatinga) in the Brazilian Semiarid. *Agricultural*
604 *and Forest Meteorology*, 287, March, 10.1016/j.agrformet.2020.107957

605 Maseyk, K., Lin, T., Rotenberg, E., Grunzweig, J.M., Schwarts, A., Yakir, D. (2008) Physiology –
606 phenology interactions in a productive semi-arid pine forest. New Phytologist, 178, 3, 603–616.

607 McNaughton, K., Jarvis, P. (1983). Predicting effects of vegetation changes on transpiration and
608 evaporation. *Water deficits and plant growth*, 7, 1-47.

609 Mello, C.R., Rodrigues, A.F., Terra, M.C.N., Mantovani, V.A., Junqueira Junior, J.A., Mello, J.M., 2023.
610 The ecohydrological function of the tropical forest rainfall interception: observation and modeling. In:
611 Saeid, Eslamian, Eslamian, Faezeh (Eds.), Handbook of HydroInformatics Volume II: Advanced Machine
612 Learning Techniques, vol. 2. Elsevier, 1ed.Amsterdan, pp. 77–103 (Org.).

613 MelMello, C.R., Avila, L.F., Lin, H., Terra, M.C.N.S., Chappell, N. (2019) Water balance in a neotropical
614 forest catchment of southeastern Brazil. *Catena*, 173, 9–21.

615 Monteith, J. L. (1965) Evaporation and environment. *Symposia of the Society for Experimental Biology*,
616 19, 205–234.

617 Muñoz-Villers, L.E., Holwerda, F., Gómez-Cárdenas, M., Equihua, M., Asbjornsen, H., Bruijnzeel, L.A.,
618 Marín-Castro, B.E., Tóbon, C. (2012) Water balances of old-growth and regenerating montane cloud forests
619 in central Veracruz, Mexico. *Journal of Hydrology*, 462–463, 53–66.

620 Myers, M., Mittermeier, R.A., Mittermeier, C.G., Fonseca, G.A.B., Kent, J. (2000) Biodiversity hotspots
621 for conservation priorities. *Nature*, 403, 2, 853–858.

Negrón Juárez, R.I., Hodnett, M.G., Fu, R., Goulden, M., von Randow, C. (2007) Control of dry season evapotranspiration over the Amazonian forest as inferred from observation at a Southern Amazon forest site. *Journal of Climate*, 20, 12, 2827–2839.

Norman, J.M., Kustas, W.P., Humes, K.S. (1995) Source approach for estimating soil and vegetation energy fluxes in observations of directional radiometric surface temperature. *Agricultural and Forest Meteorology*, 77, 3–4, 263–293.

Oliveira-Filho, A.T., Scolforo, J.R.S., Mello, J.M. (1994) Composição florística e estrutura comunitária de um remanescente de floresta semidecídua montana em Lavras, MG. *Rev. Bras. Bot.*, 17, 167-182.

Pereira, D.R., Mello, C.R., Silva, A.M., Yanagi, S.N.M. (2010) Evapotranspiration and estimation of aerodynamic and stomatal conductance in a fragment of Atlantic Forest in Mantiqueira Range Region, MG. *Cerne*, 16, 32–40.

Poorter, L., Bongers, F., Aide, T.M. et al. (2016) Biomass resilience of Neotropical secondary forests. *Nature*, 530, 7589, 211–214.

Priestley, C.H.B., Taylor, R.J. (1972) On the Assessment of Surface Heat Flux and Evaporation Using Large-Scale Parameters. *Monthly Weather Review,* 100, 2, 81–92.

Qiu, R., Liu, C., Cui, N., Wu, Y., Zhenchang, W. (2019). Evapotranspiration estimation using a modified Priestley-Taylor model in a rice-wheat rotation system. *Agricultural Water Management,* 224, 105755

Rezende, C.L., Scarano, F.R., Assad, E.D., Joly, C.A., Metzger, J.P., Strassburg, B.B.N., Tabarelli, M., Fonseca, G.A., Mittermeier, R.A. (2018) From hotspot to hopespot: An opportunity for the Brazilian Atlantic Forest. *Perspectives in Ecology and Conservation,* 16, 4, 208–214.

Rocha, H.R., Goulden, M.L., Miller, S.D., Menton, M.C., Pinto, L.D.V.O., Freitas, P.H.C., Figueira, A.M.S. (2004). Seasonality of water and heat fluxes over a tropical forest in eastern Amazonia. *Ecological Applications*, 14, 4, 22–32.

Rodrigues, A.F., Mello, C.R., Nehren, U., Ribeiro, J.P.C., Mantovani, V.A., Mello, J.M. (2021). Modeling canopy interception under drought conditions: The relevance of evaporation and extra sources of energy, *Journal of Environmental Management,* 292, 112710.

Rodrigues, A.F., Mello, C.R., Terra, M.D.C.N.S., Silva, V.O., Pereira, G.A. (2020). Soil water content and net precipitation spatial variability in an Atlantic forest remnant. *Acta Sci. Agron.* 42, 1–13.

Salemi, L.F., Groppo, J., Trevisan, R. et al. (2013) Land-use change in the Atlantic rainforest region: Consequences for the hydrology of small catchments. *Journal of Hydrology*, 499, 100–109.

Scarano, F.R., Ceotto, P. (2015) Brazilian Atlantic forest: impact, vulnerability, and adaptation to climate change. *Biodiversity and Conservation*, 24, 9, 2319–2331.

Silberstein, R.P., Held, A., Hatton, T., Viney, N., Sivapalan, M. (2003). Modelling the energy balance of a natural jarrah (Eucalyptus marginata) forest. *Agricultural and Forest Meteorology,* 115, 3–4, 201–230.

Su, Z. (2002) The Surface Energy Balance System (SEBS) for estimation of turbulent heat fluxes. *Hydrology and Earth System Sciences*, 6, 1, 85–99.

Taffarello, D., Calijuri, M.C., Viani, R.A.G., Marengo, J.A., Mediondo, E.M. (2017). Hydrological services in the Atlantic Forest, Brazil: An ecosystem-based adaptation using ecohydrological monitoring. Climate Services, 8, December, 1-16.

Tambosi, L.R., Martensen, A.C., Ribeiro, M.C., Metzger, J.P. (2014) A framework to optimize biodiversity restoration efforts based on habitat amount and landscape connectivity. *Restoration Ecology,* 22, 2, 169–177.

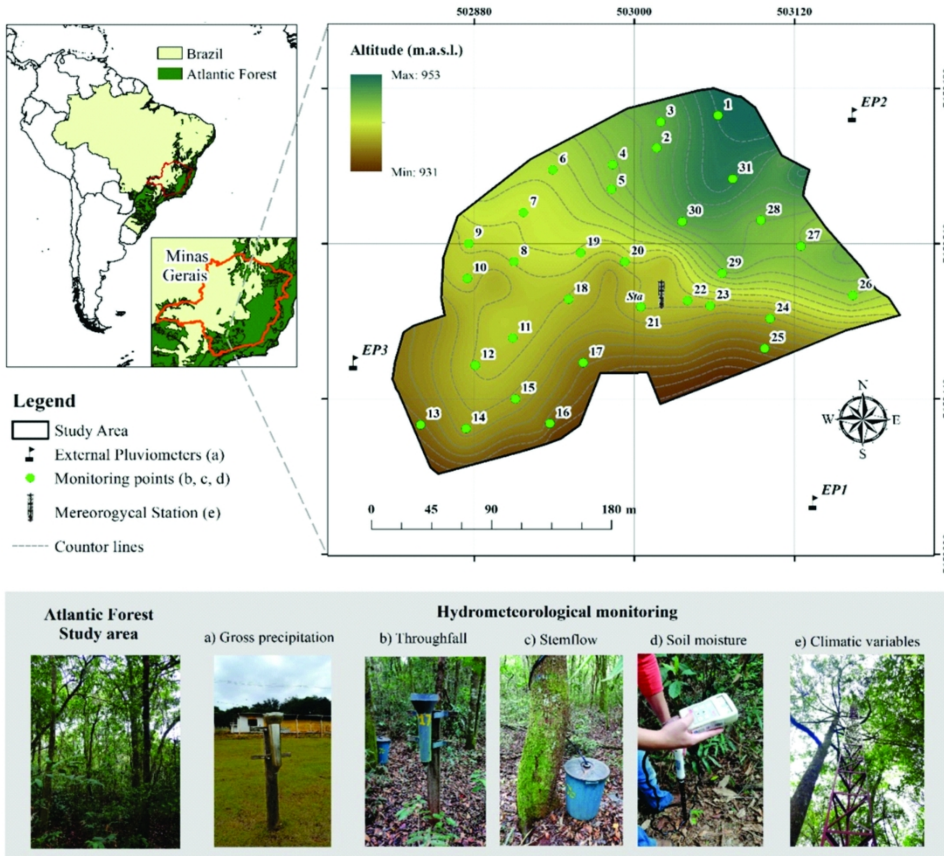
Tan, Z.H., Zhao, J.F., Wang, G.Z. et al. (2019) Surface conductance for evapotranspiration of tropical forests: Calculations, variations, and controls. *Agricultural and Forest Meteorology,* 275, June, 317–328. 10.1016/j.agrformet.2019.06.006

Terra, M.C.N.S., Mello, C.R., Mello, J.M., Oliveira, V.A., Nunes, M.H., Silva, V.O., Rodrigues, A.F., Alves, G.J., 2018. Stemflow in a neotropical forest remnant: vegetative determinants, spatial distribution and correlation with soil moisture. Trees (Berl.) 32, 323–335.

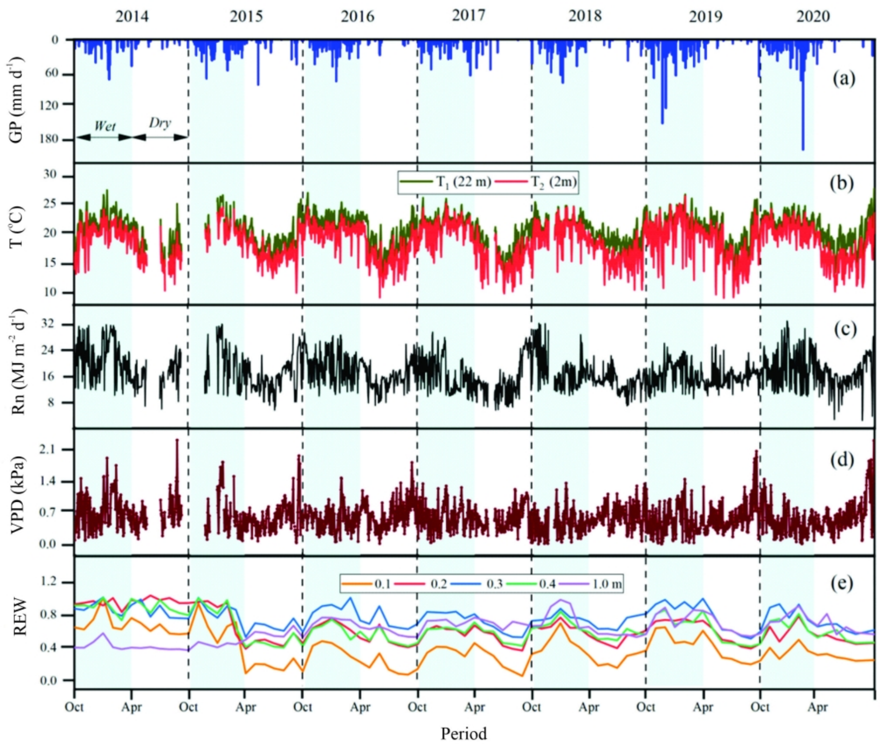
Tomasella, J., Hodnett, M.G., Cuartas, L.A., Nobre, A.D., Waterloo, M.J., Oliveira, S.M. (2008) The water balance of an Amazonian micro-catchment: the effect of interannual variability of rainfall on hydrological behaviour. *Hydrological Processes*, 2274, 2267–2274.

Von Randow, R.C.S., Tomasella, J., von Randow, C., Araújo, A.C., Manzi, A.O., Hutjes, R., Kruijt, B. (2020) Evapotranspiration and gross primary productivity of secondary vegetation in Amazonia inferred by eddy covariance. *Agricultural and Forest Meteorology*, 294, 108141.

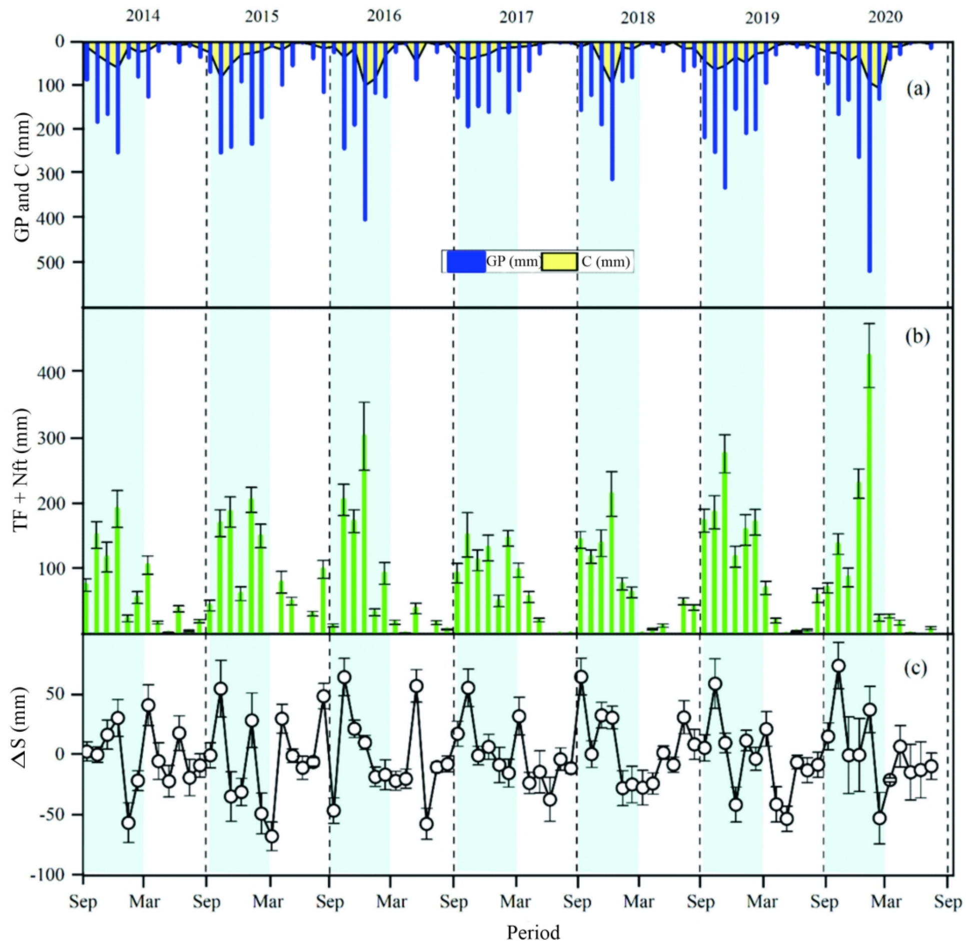
Zanini, A.M., Mayrinck, R.C, Viera, S.A., Camargo, P.B., Rodrigues, R.R. (2021) The effect of ecological restoration methods on carbon stocks in the Brazilian Atlantic Forest. *Forest Ecology and Management*, 481, 118734.



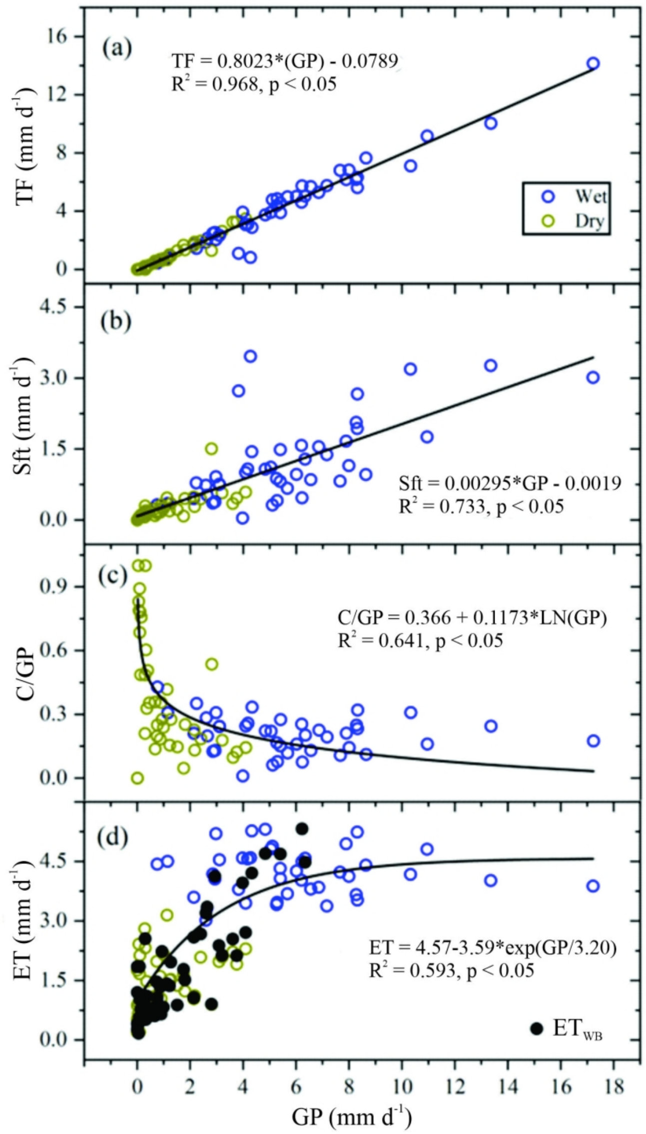
226x212mm (300 x 300 DPI)



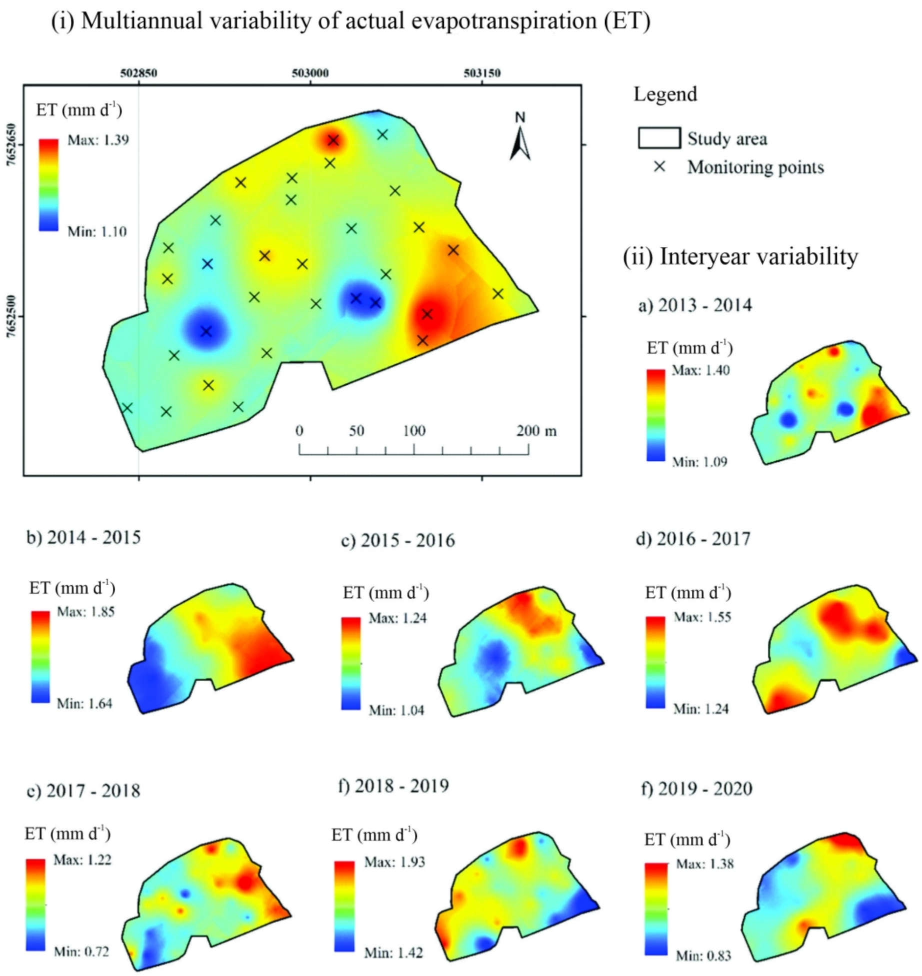
237x181mm (300 x 300 DPI)



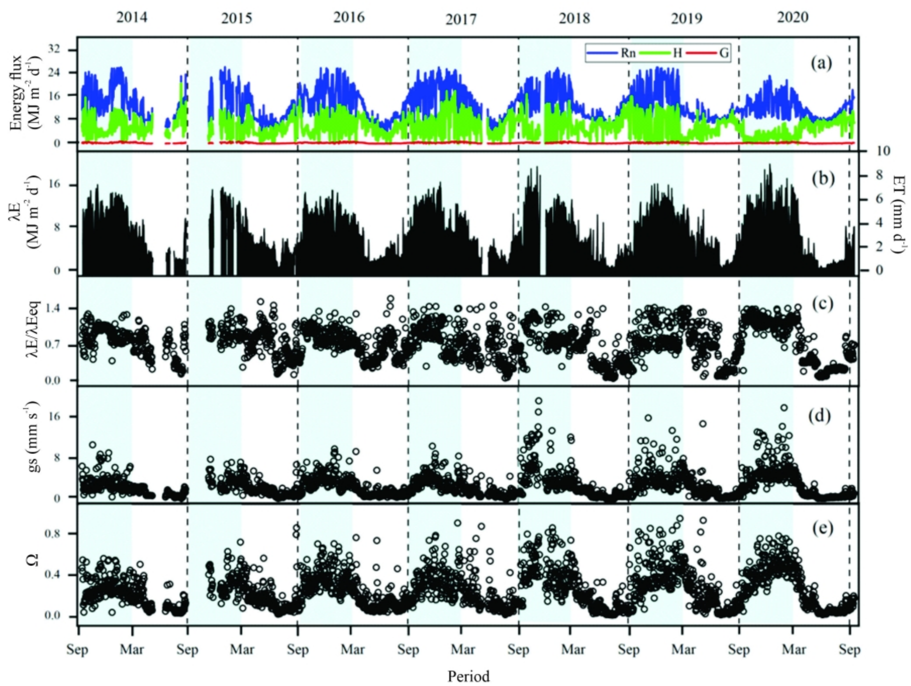
205x196mm (300 x 300 DPI)



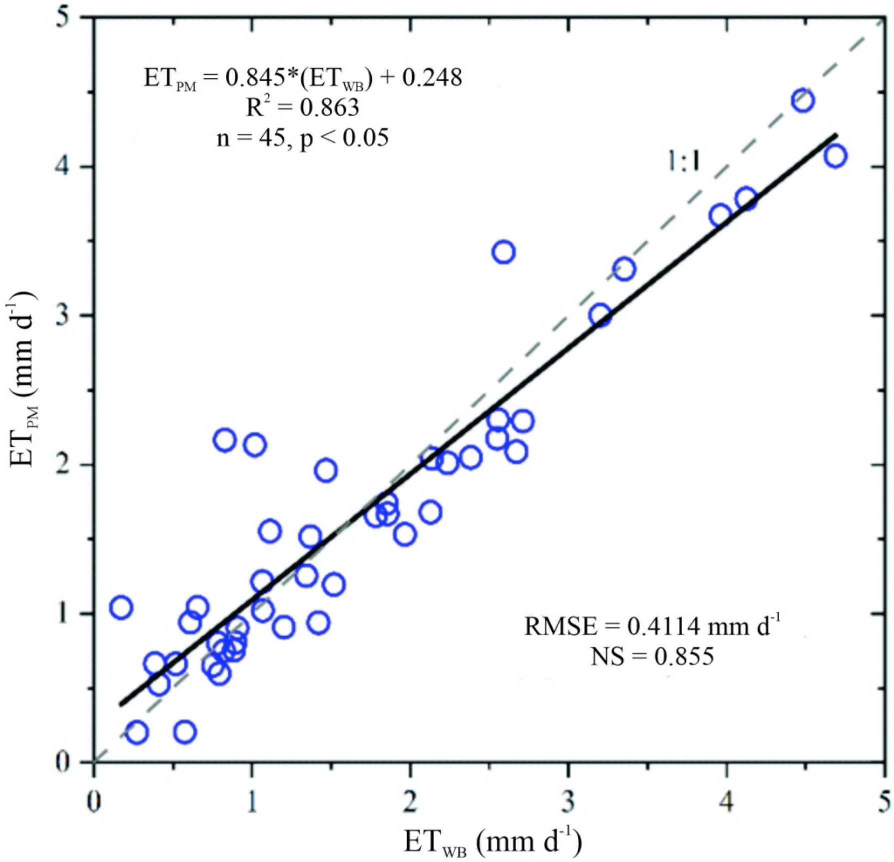
122x204mm (300 x 300 DPI)



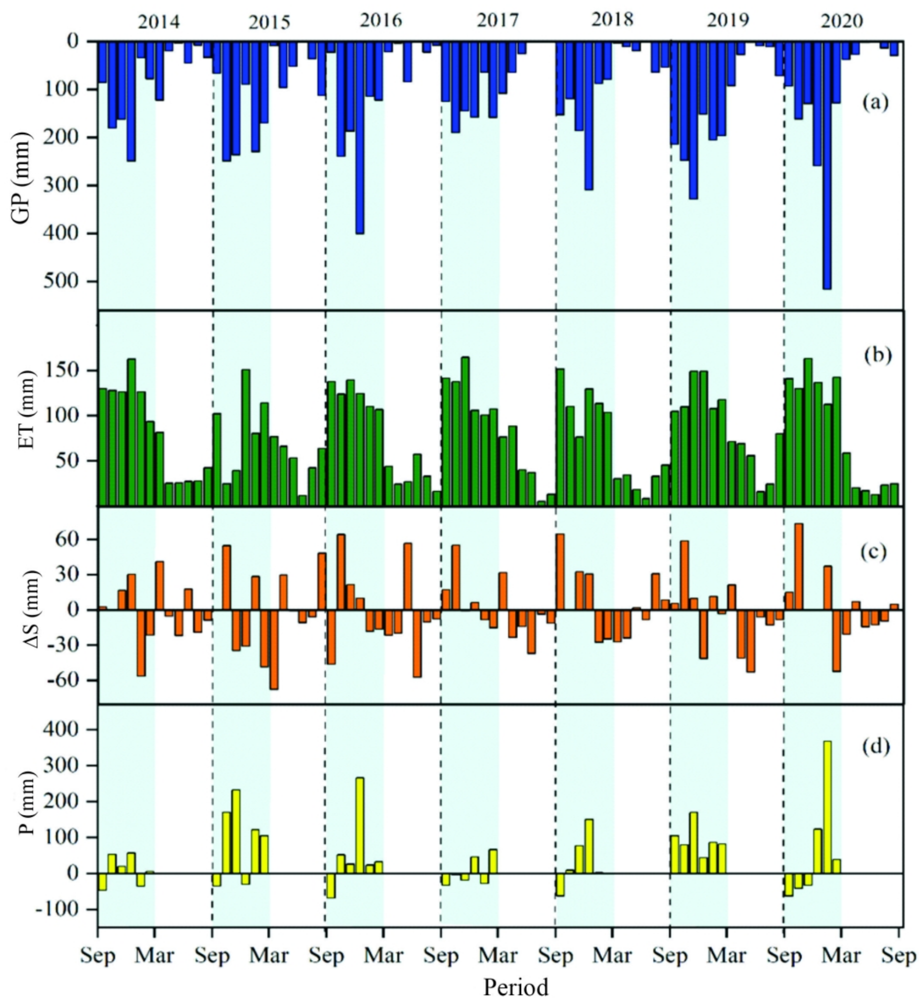
197x207mm (300 x 300 DPI)



260x181mm (300 x 300 DPI)



155x148mm (300 x 300 DPI)



196x204mm (300 x 300 DPI)

Supplementary

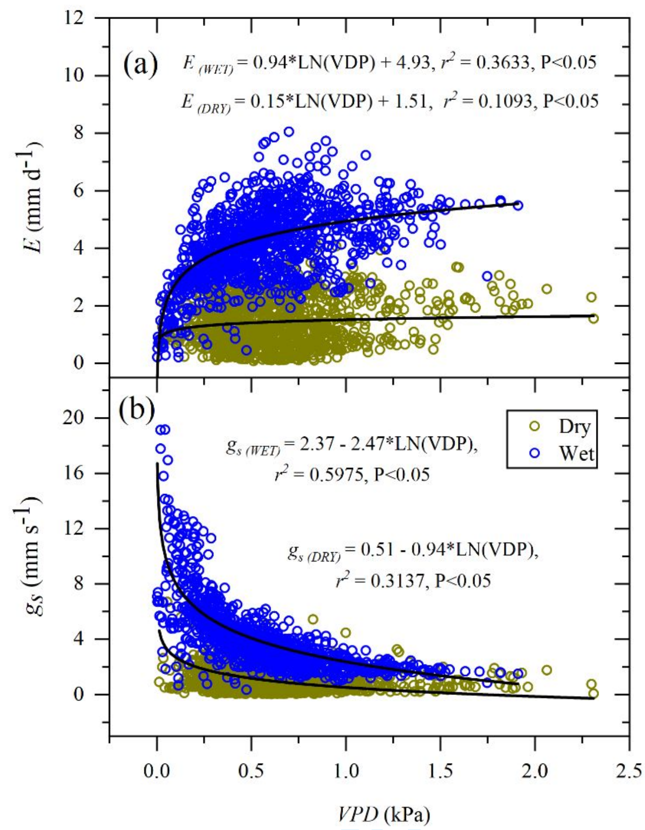


Figure 1S. Relationships between vapor pressure deficit (VPD) with (a) daily modeled evapotranspiration and (b) daily g_s.

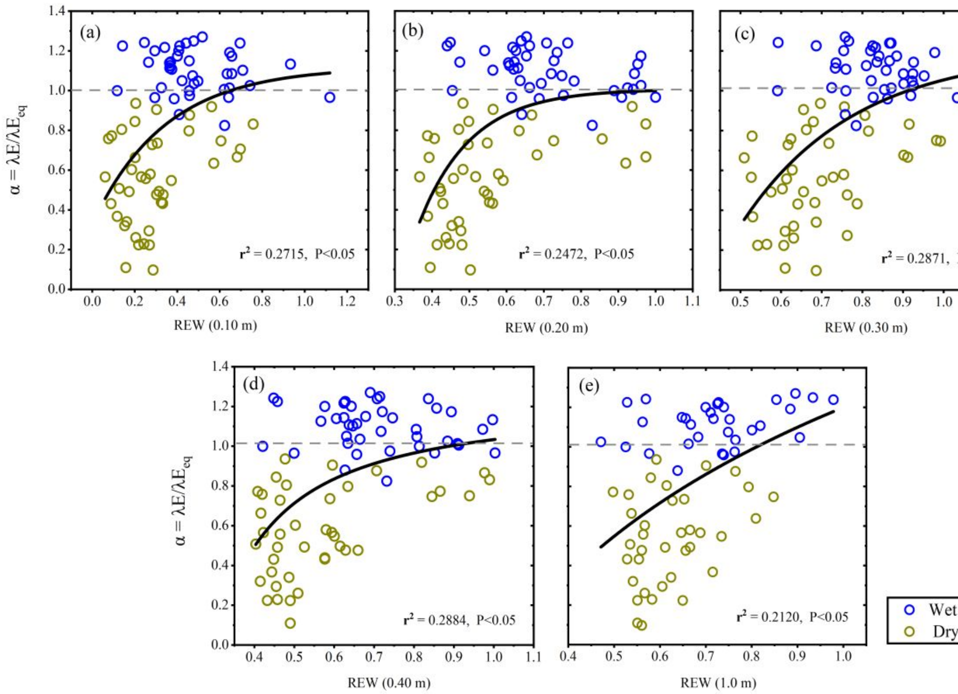


Figure 2S. Relationships between the monthly mean values of the Priestley-Taylor coefficient ($\alpha = \lambda E/\lambda E_{eq}$) and the relative extractable soil water content (REW) (a. 0.10 m; b. 0.20 m; c. 0.30 m; d. 0.40 m; and e. 1.0 m).

Table A1. Daily average of gross precipitation (GP, mm d^{-1}), throughfall (TF, mm d^{-1}), streamflow (St), and canopy rainfall interception (C) in the dry and wet periods of the hydrological years from 2014 to 2020.

Hydrological Year		GP ($mm.d^{-1}$)	TF ($mm.d^{-1}$)	St ($mm.d^{-1}$)	C ($mm.d^{-1}$)
2013-2014	Wet	4.13 ± 2.7	3.19 ± 2.1	0.01 ± 0.02	0.85 ± 0.6
	Dry	0.9 ± 1.5	0.62 ± 1.3	0.01 ± 0.01	0.18 ± 0.2
	Annual	2.51 ± 2.1	1.9 ± 1.7	0.01 ± 0.01	0.52 ± 0.4
2014-2015	Wet	6.67 ± 2.6	5.3 ± 2.2	0.02 ± 0.02	0.87 ± 0.8
	Dry	1.49 ± 1.5	1.34 ± 1.4	0.01 ± 0.01	0.24 ± 0.2
	Annual	4.08 ± 2.1	3.32 ± 1.8	0.01 ± 0.01	0.56 ± 0.5
2015-2016	Wet	5.16 ± 4.3	4.41 ± 3.7	0.01 ± 0.02	1.08 ± 1.2
	Dry	0.51 ± 1.0	0.4 ± 0.51	0.01 ± 0.01	0.13 ± 0.6
	Annual	2.84 ± 2.7	2.4 ± 2.1	0.01 ± 0.01	0.61 ± 0.9
2016-2017	Wet	5.05 ± 1.4	4.07 ± 1.3	0.01 ± 0.01	0.98 ± 0.4
	Dry	0.46 ± 1.5	0.36 ± 1.3	0.01 ± 0.00	0.12 ± 0.1
	Annual	2.76 ± 1.5	2.21 ± 1.3	0.01 ± 0.00	0.54 ± 0.3
2017-2018	Wet	4.54 ± 2.8	4.28 ± 1.8	0.01 ± 0.01	0.45 ± 1.2
	Dry	0.51 ± 0.9	0.33 ± 0.7	0.01 ± 0.00	0.18 ± 0.2
	Annual	2.53 ± 1.9	2.3 ± 1.3	0.01 ± 0.00	0.31 ± 0.7
2018-2019	Wet	7.01 ± 2.0	5.73 ± 1.7	0.01 ± 0.01	1.47 ± 0.4
	Dry	0.66 ± 1.2	0.43 ± 1.0	0.01 ± 0.00	0.23 ± 0.2
	Annual	3.83 ± 1.6	3.08 ± 1.4	0.01 ± 0.00	0.85 ± 0.3
2019-2020	Wet	4.86 ± 5.3	3.73 ± 4.9	0.01 ± 0.02	1.20 ± 1.2
	Dry	0.68 ± 0.5	0.44 ± 0.4	0.01 ± 0.01	0.21 ± 0.1
	Annual	2.77 ± 2.9	2.08 ± 2.6	0.01 ± 0.01	0.72 ± 0.7

Table A2. Daily average values of net radiation (R_n, MJ m^{-2} d^{-1}), latent heat flux (LE, MJ m^{-2} d^{-1}), sensible heat flux (H, MJ m^{-2} d^{-1}), soil heat flux (G, MJ m^{-2} d^{-1}), latent heat flux ($\lambda E_{res,}$ MJ m^{-2} d^{-1}), coefficient of Priestley-Taylor ($\alpha = \lambda E/\lambda E_{eq}$), canopy conductance ($g_s$, mm s^{-1}), decoupling factor (Ω) and evapotranspiration (E, mm d^{-1}) in the dry, wet and annual time scales of the hydrological years from 2014 to 2020.

Hydrological Year		R_n	H	G	λE_{res}	A	g_s	Ω	E
2013-2014	Wet	15.89 ± 5.1	5.38 ± 3.1	0.45 ± 0.2	10.64 ± 2.9	0.89 ± 0.2	2.29 ± 1.6	0.25 ± 0.1	4.33 ± 1.2
	Dry	9.44 ± 3.3	5.62 ± 3.1	0.2 ± 0.1	3.26 ± 2.2	0.57 ± 0.3	0.69 ± 0.7	0.09 ± 0.1	1.33 ± 0.9
	Annual	12.67 ± 4.2	5.5 ± 3.1	0.32 ± 0.1	6.95 ± 2.5	0.73 ± 0.2	1.49 ± 1.2	0.17 ± 0.1	2.83 ± 1.0
2014-2015	Wet	17.48 ± 5.9	6.09 ± 3	0.47 ± 0.2	10.49 ± 3.4	0.9 ± 0.2	3.39 ± 1.5	0.31 ± 0.1	4.29 ± 1.4
	Dry	8.61 ± 3.8	4.95 ± 3.2	0.17 ± 0.1	4.3 ± 1.60	0.63 ± 0.3	1.34 ± 1	0.14 ± 0.1	1.75 ± 0.7
	Annual	13.04 ± 4.9	5.52 ± 3.1	0.32 ± 0.1	7.39 ± 2.5	0.77 ± 0.2	2.36 ± 1.2	0.22 ± 0.1	3.02 ± 1.0
2015-2016	Wet	16.95 ± 4.7	5.6 ± 3.1	0.42 ± 0.2	10.05 ± 2.3	0.86 ± 0.2	3.34 ± 1.5	0.33 ± 0.1	4.1 ± 0.90
	Dry	8.87 ± 3.8	6.26 ± 2.6	0.17 ± 0.1	2.82 ± 2.1	0.55 ± 0.2	0.88 ± 0.6	0.11 ± 0.1	1.15 ± 0.8
	Annual	12.91 ± 4.3	5.93 ± 2.8	0.29 ± 0.1	6.43 ± 2.2	0.7 ± 0.2	2.11 ± 1.1	0.22 ± 0.1	2.63 ± 0.9
2016-2017	Wet	18.71 ± 5.1	7.06 ± 4.2	0.45 ± 0.2	9.63 ± 3.0	0.87 ± 0.3	3.17 ± 1.6	0.31 ± 0.1	3.93 ± 1.3
	Dry	10.1 ± 3.9	7.24 ± 3.2	0.19 ± 0.1	3.44 ± 2.1	0.57 ± 0.3	1.04 ± 1.0	0.12 ± 0.1	1.4 ± 0.8
	Annual	14.4 ± 4.5	7.15 ± 3.7	0.32 ± 0.1	6.53 ± 2.6	0.72 ± 0.3	2.11 ± 1.3	0.22 ± 0.1	2.67 ± 1.0
2017-2018	Wet	15.56 ± 4.7	5.53 ± 3.6	0.45 ± 0.2	10.42 ± 3.3	0.84 ± 0.2	3.75 ± 3.7	0.41 ± 0.2	4.26 ± 1.4
	Dry	9.08 ± 3.2	6.51 ± 3.1	0.17 ± 0.1	2.29 ± 2.0	0.33 ± 0.3	0.6 ± 0.7	0.07 ± 0.1	0.94 ± 0.8
	Annual	12.32 ± 3.9	6.02 ± 3.4	0.31 ± 0.1	6.36 ± 2.7	0.59 ± 0.3	2.18 ± 2.2	0.24 ± 0.1	2.6 ± 1.10
2018-2019	Wet	15.82 ± 5.9	5.76 ± 4.3	0.4 ± 0.2	10.1 ± 3.0	0.81 ± 0.3	3.62 ± 2.4	0.38 ± 0.2	4.13 ± 1.3
	Dry	9.07 ± 1.5	5.33 ± 2.3	0.18 ± 0.1	3.78 ± 2.3	0.62 ± 0.3	0.91 ± 1.1	0.12 ± 0.1	1.54 ± 0.9
	Annual	12.45 ± 3.7	5.54 ± 3.3	0.29 ± 0.1	6.94 ± 2.6	0.71 ± 0.3	2.27 ± 1.7	0.25 ± 0.2	2.83 ± 1.1
2019-2020	Wet	13.1 ± 3.5	2.2 ± 1.3	0.34 ± 0.1	10.66 ± 3.1	1.15 ± 0.1	4.71 ± 2.7	0.45 ± 0.1	4.35 ± 1.3
	Dry	9.36 ± 2.7	7.22 ± 1.7	0.18 ± 0.1	1.91 ± 1.9	0.27 ± 0.2	0.48 ± 0.6	0.06 ± 0.1	0.78 ± 0.8
	Annual	11.23 ± 3.1	4.71 ± 1.5	0.26 ± 0.1	6.29 ± 2.5	0.71 ± 0.2	2.59 ± 1.6	0.25 ± 0.1	2.57 ± 1.0